Horizon EU HealthRiskADAPT: A focus on tool development to reduce indoor air pollution exposure. And why it matters

Federico Dallo

ISGlobal Seminar, 21 November 2025, Campus Mar - Barcelona

Introduction

Consiglio Nazionale delle Ricerche

Who am I

- Chemist
- PhD in Environmental Science
- Marie-Curie Postdoctoral Fellow
- Researcher at ISP-CNR

About what I'm doing

- Tools for Environmental Monitoring
- Modeling and Monitoring Indoors
- → Health Impact Assessment (would like to learn…)

Why I think this is relevant?

- Identify the right approach to influence **behavior** toward building more **resilient** and **sustainable societies** in the context of climate change

Outline

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Environmental Monitoring

- Low-Cost Sensors
- LCS Data Quality

Built Environment and Wildfire Air Pollution

- Wildfires – Case Study in CA
- Modelling Infiltration of Outdoor PM2.5 Indoors
- Reducing Indoor Exposure to PM2.5

healthRiskADAPT project

- Tools Development for Increasing Resilience to Climate Stressors

Environmental Monitoring

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Increasing the maturity of measurements of essential climate variables (ECVs) at Italian atmospheric WMO/GAW observatories by implementing automated data elaboration chains. L Naitza, P Cristofanelli, A Marinoni, F Calzolari… - Computers & Geosciences, 2020

THE QR CODE GENERATOR

Data Management

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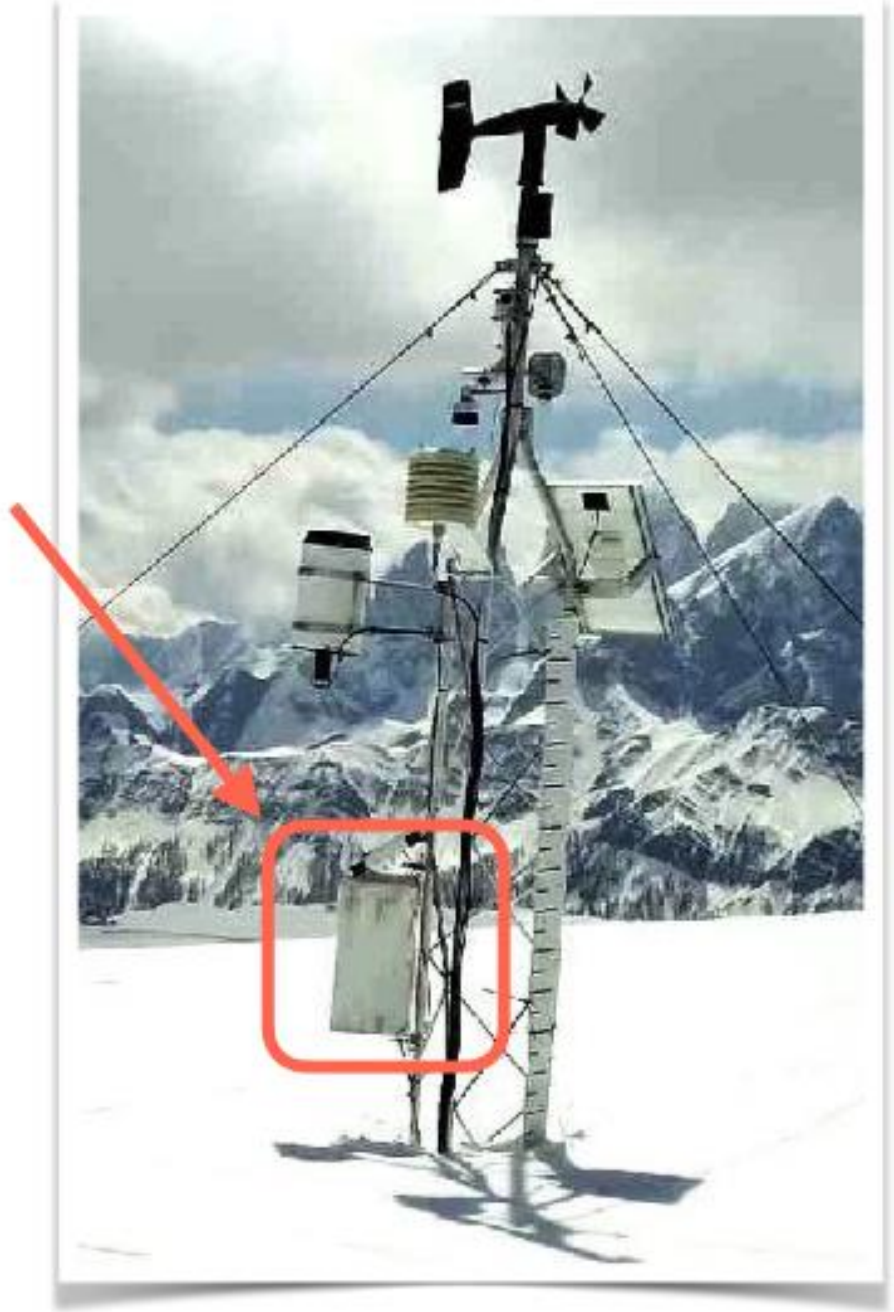
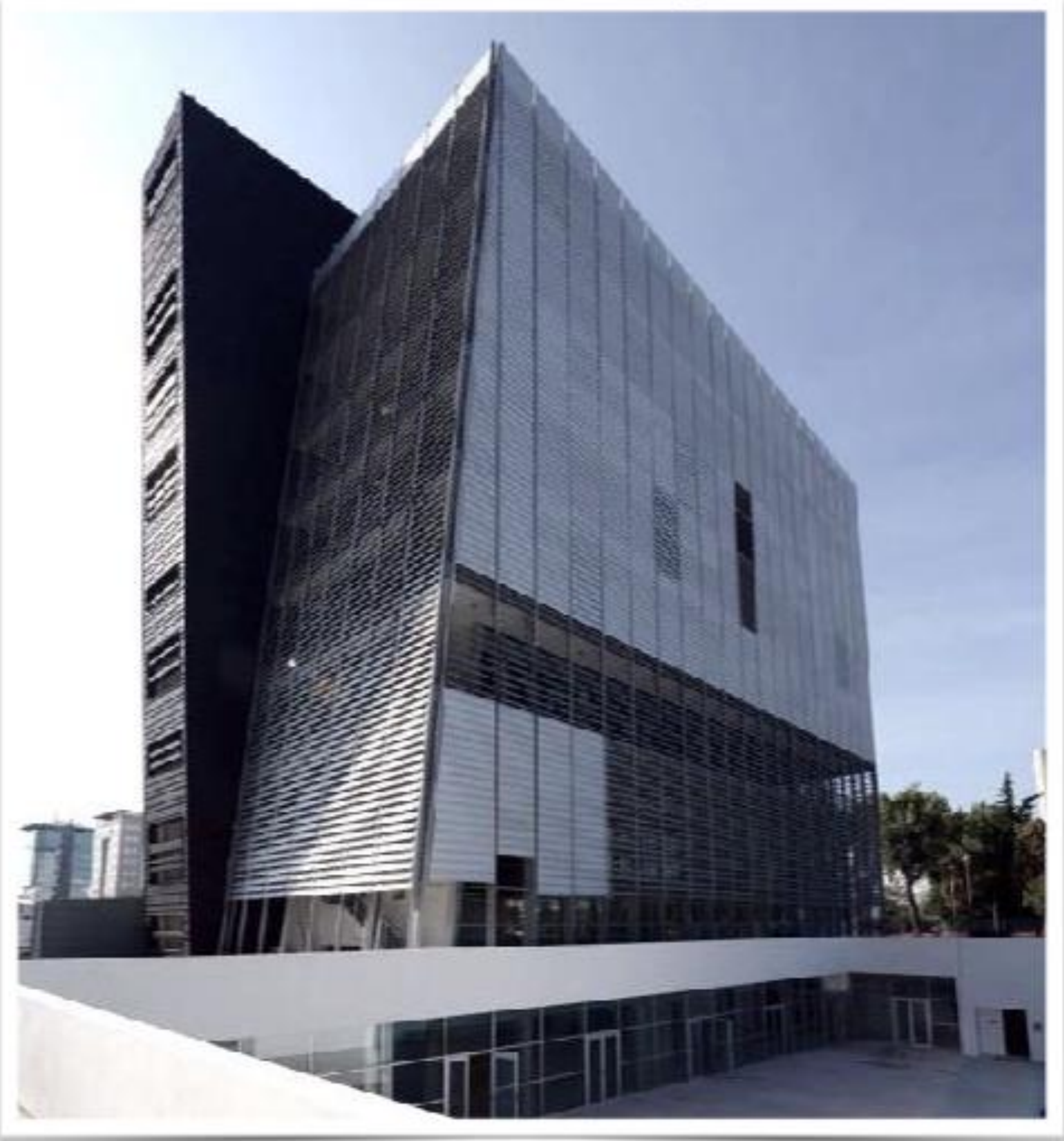
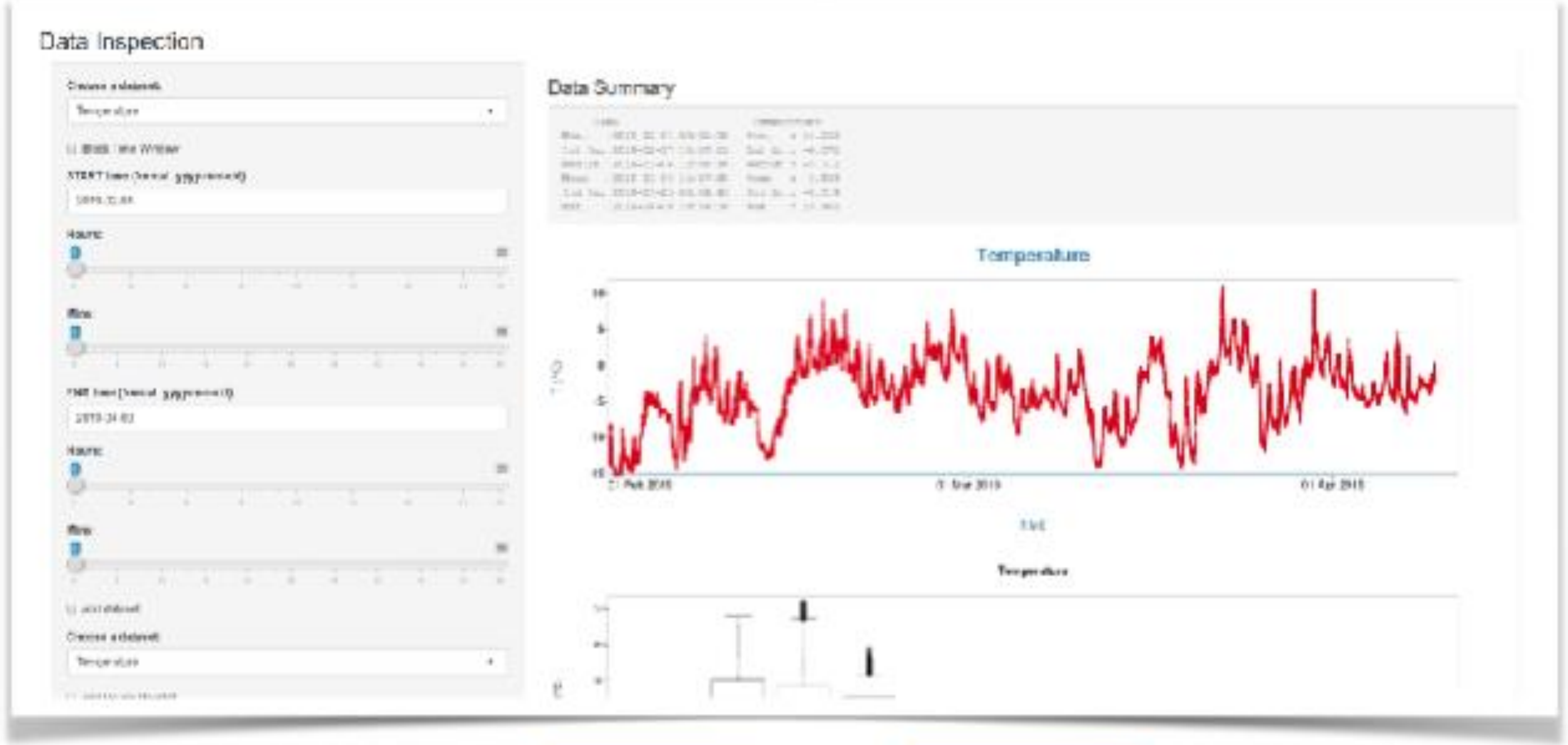
Col Margherita

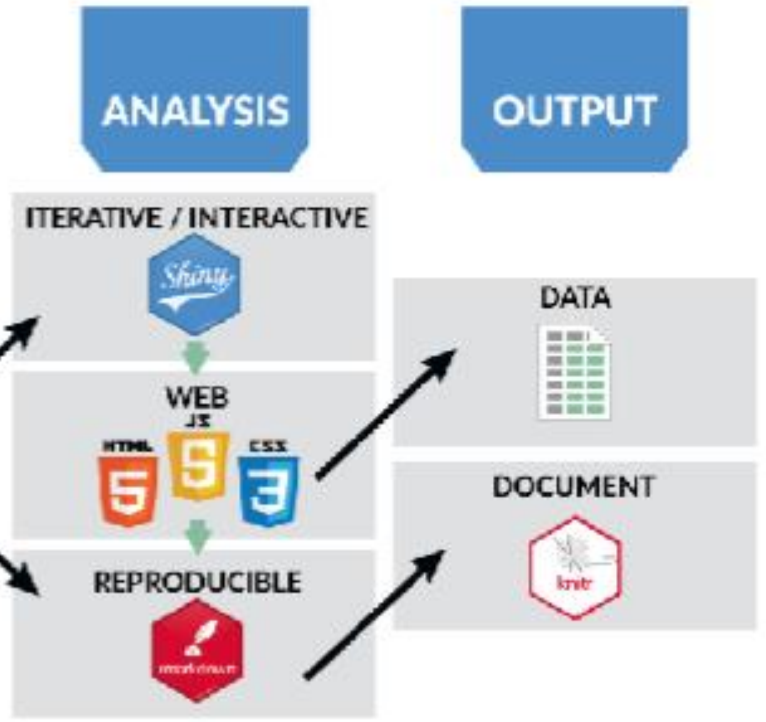


Scientific Campus
Ca' Foscari University of Venice



Applications, docs, tools, etc..





Acquisition

"Mainframe" & Database

https://github.com/theRosyProject/PIONEER

Why Low-Cost Sensors?

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Ozone and Reactive Gases are of high relevance to the Earth's climate ecosystems and human health

- Gaps remain in the surface observation networks
- Improve measurement accuracy and spatial/temporal sampling
- Polar and Alpine environment (the Climate Sentinels)
    - Reference background sites due to their sensitivity to Climate Change
    - Hard to access
    - Lack of infrastructure and cost of maintenance
    - Absence of grid power
    - Telecommunication

Assessing LCS Accuracy

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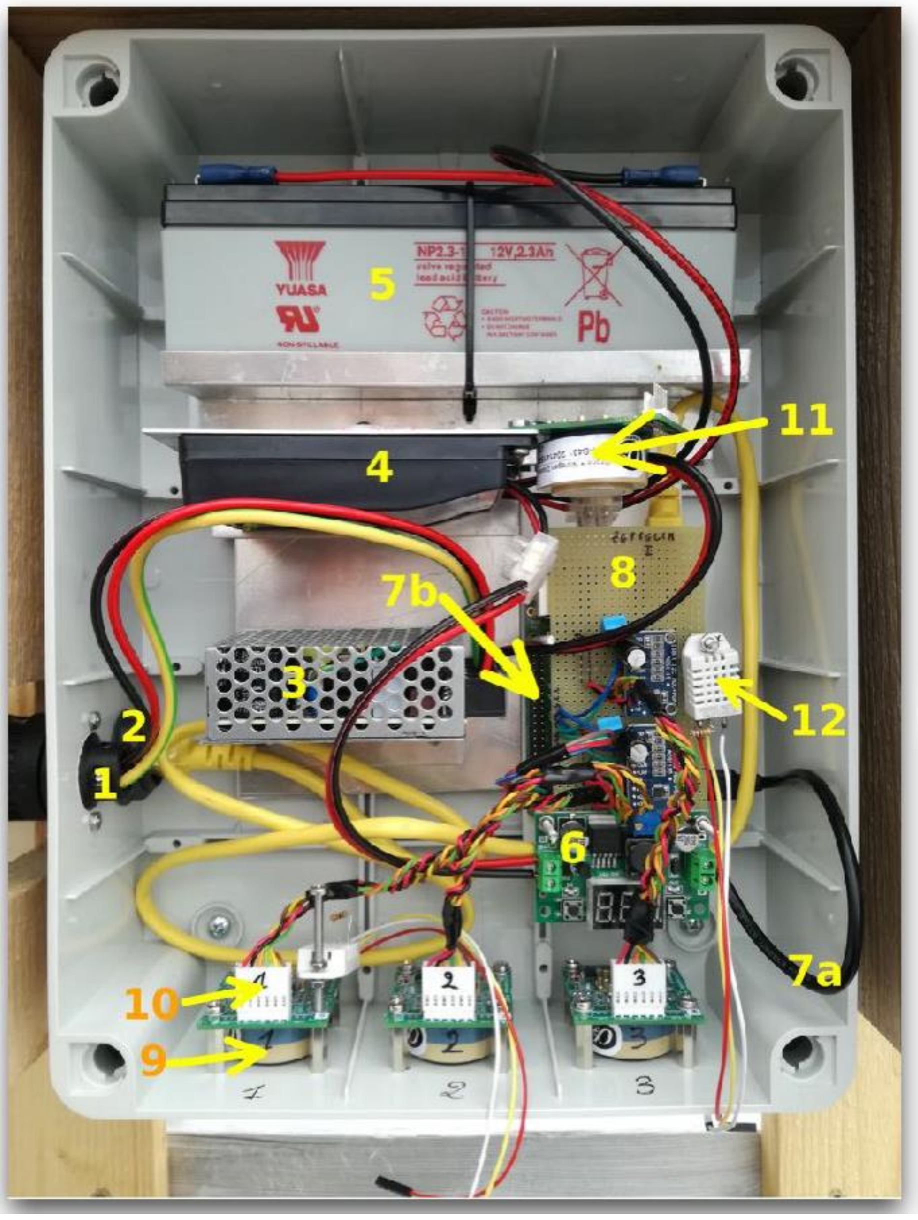
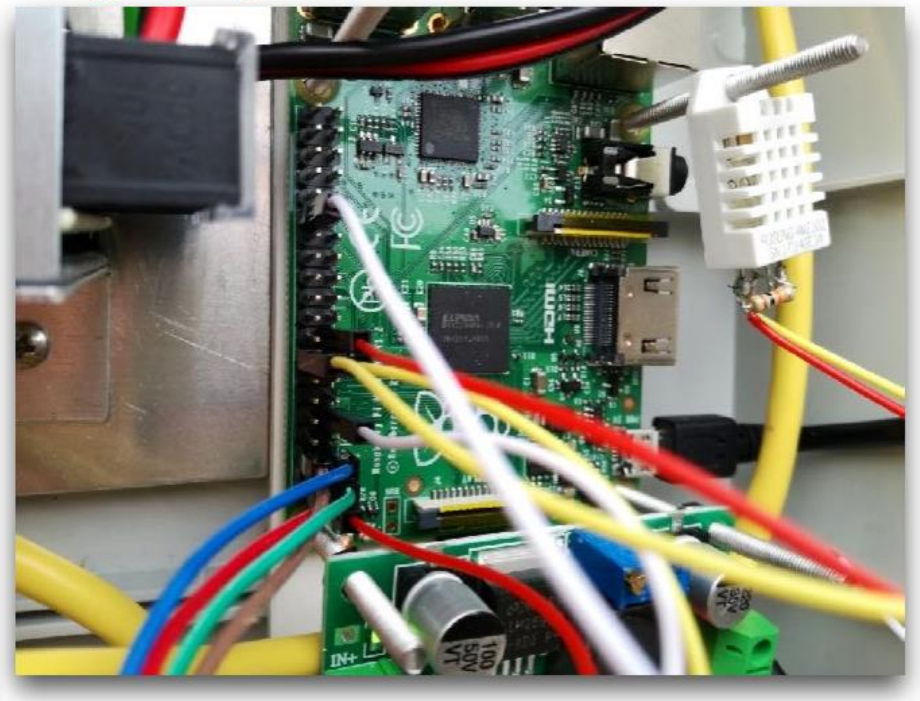
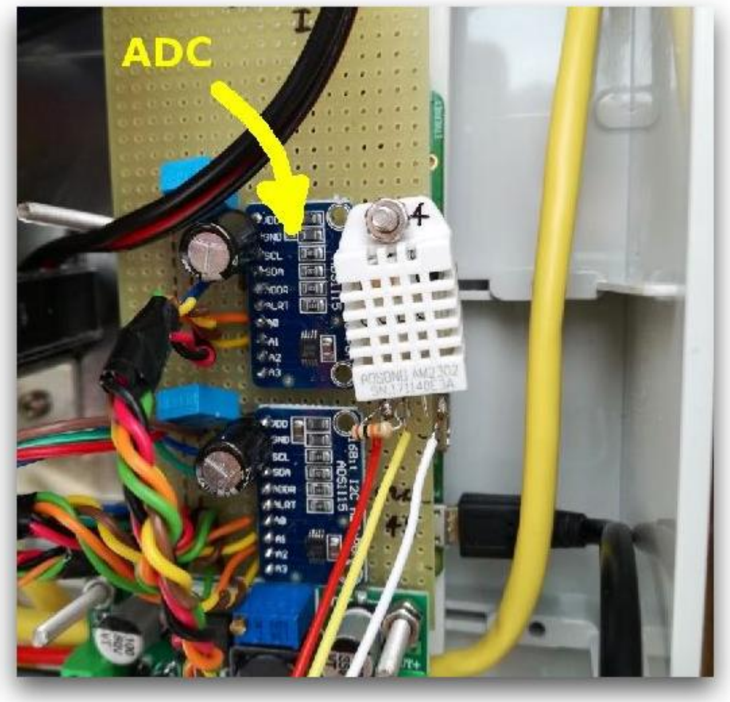
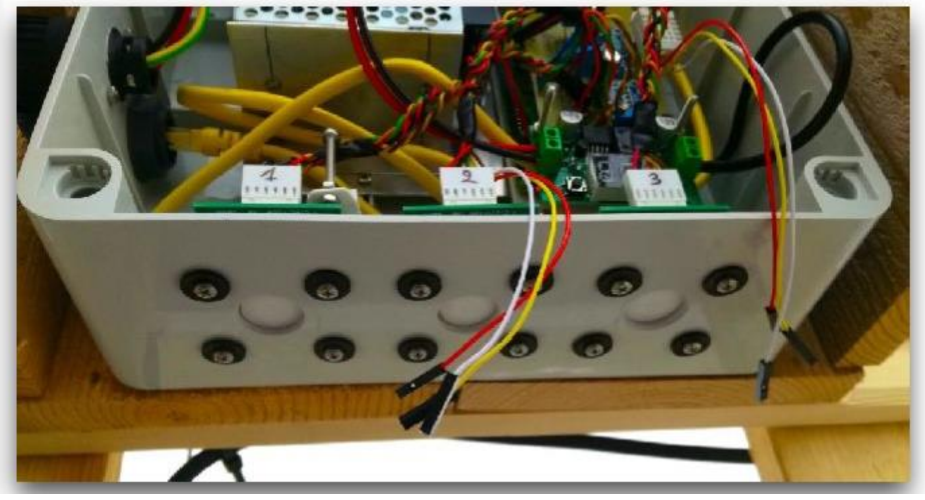
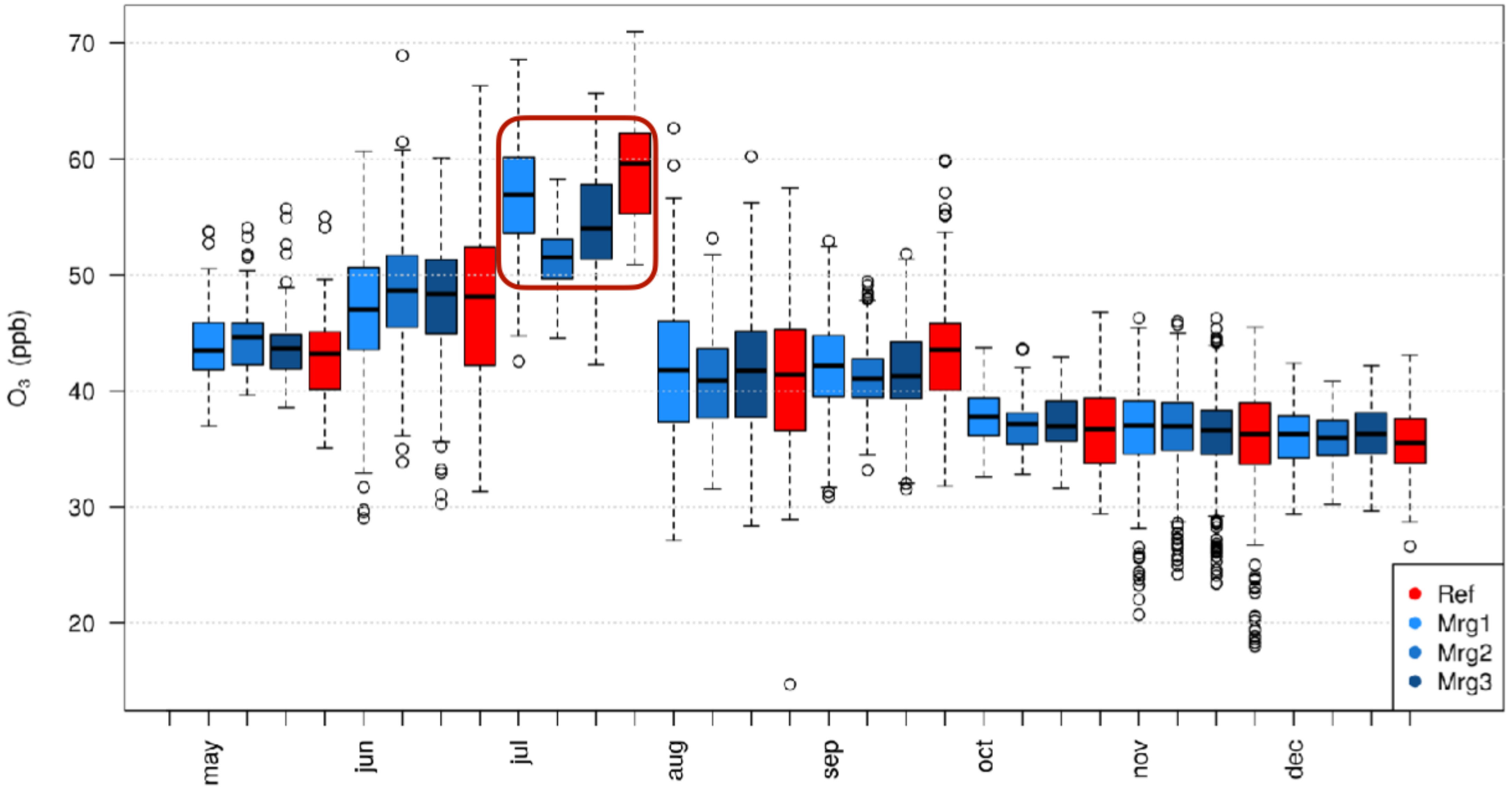








https://www.alphasense.com/products/ozone

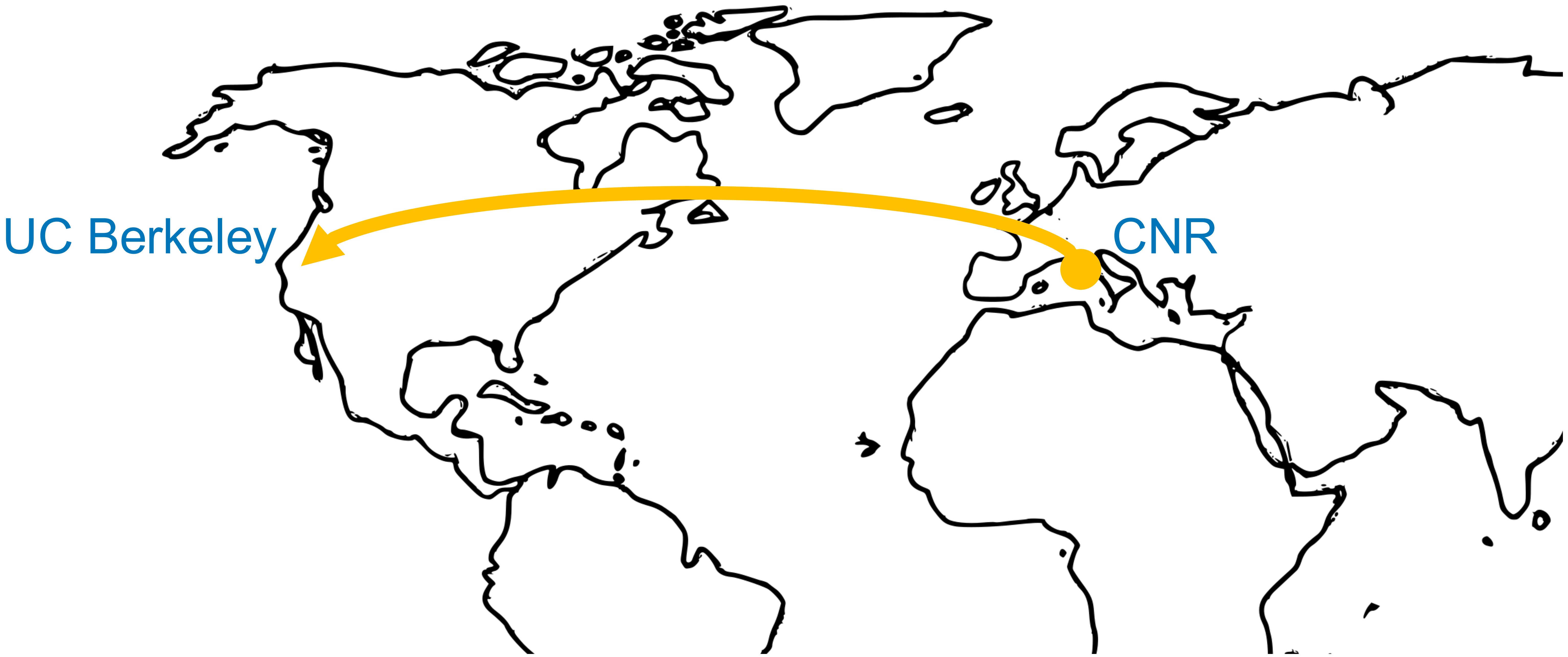


Calibration and assessment of electrochemical low-cost sensors in remote alpine harsh environments. F Dallo, D Zannoni, J Gabrieli, P Cristofanelli… - Atmospheric Measurement Techniques, 2021

LOD = 5 ppb
LOQ = 15 ppb

PCC > 0.99 up to 250 ppb in Laboratory
PCC ~ 0.8 in Field

bias ~ 6 ppb without correction
bias ~ 3.5 ppb
bias ±8.5 at 95% of confidence



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Built Environment and Wildfire Air Pollution

Background

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Wildfires

- Billion of dollars in annual losses in California
- Main source of particulate matter in west US

Fine Particulate Matter (PM2.5)

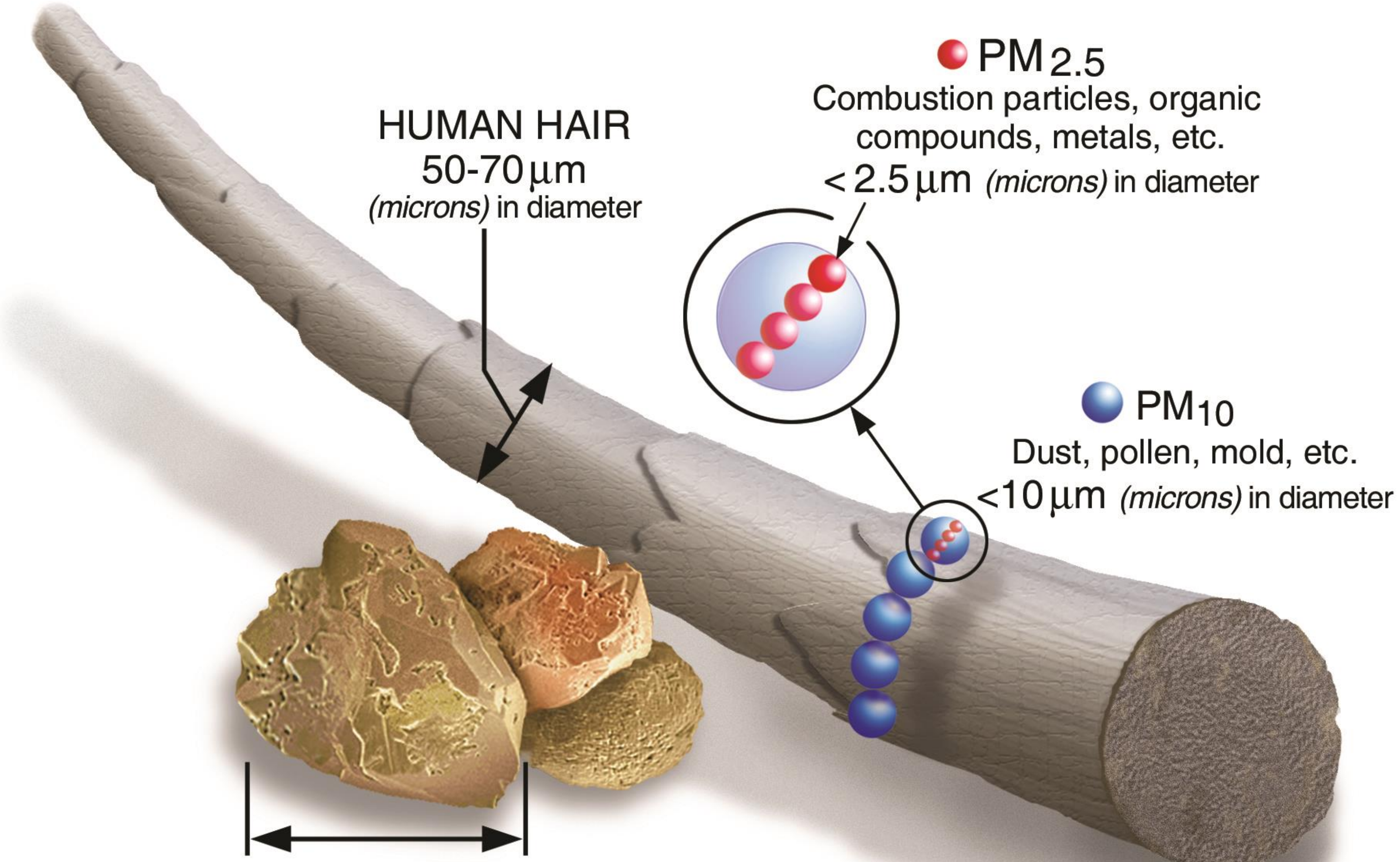
- Millions of premature death each year

Climate Change

- 27% increased risk of wildfire by 2050 relative to 2000 levels

Smoke from the 2018 Camp fire caused weeks of poor air quality in the San Francisco Bay area.
Image: EPA

Recap: why PM2.5 is a concern



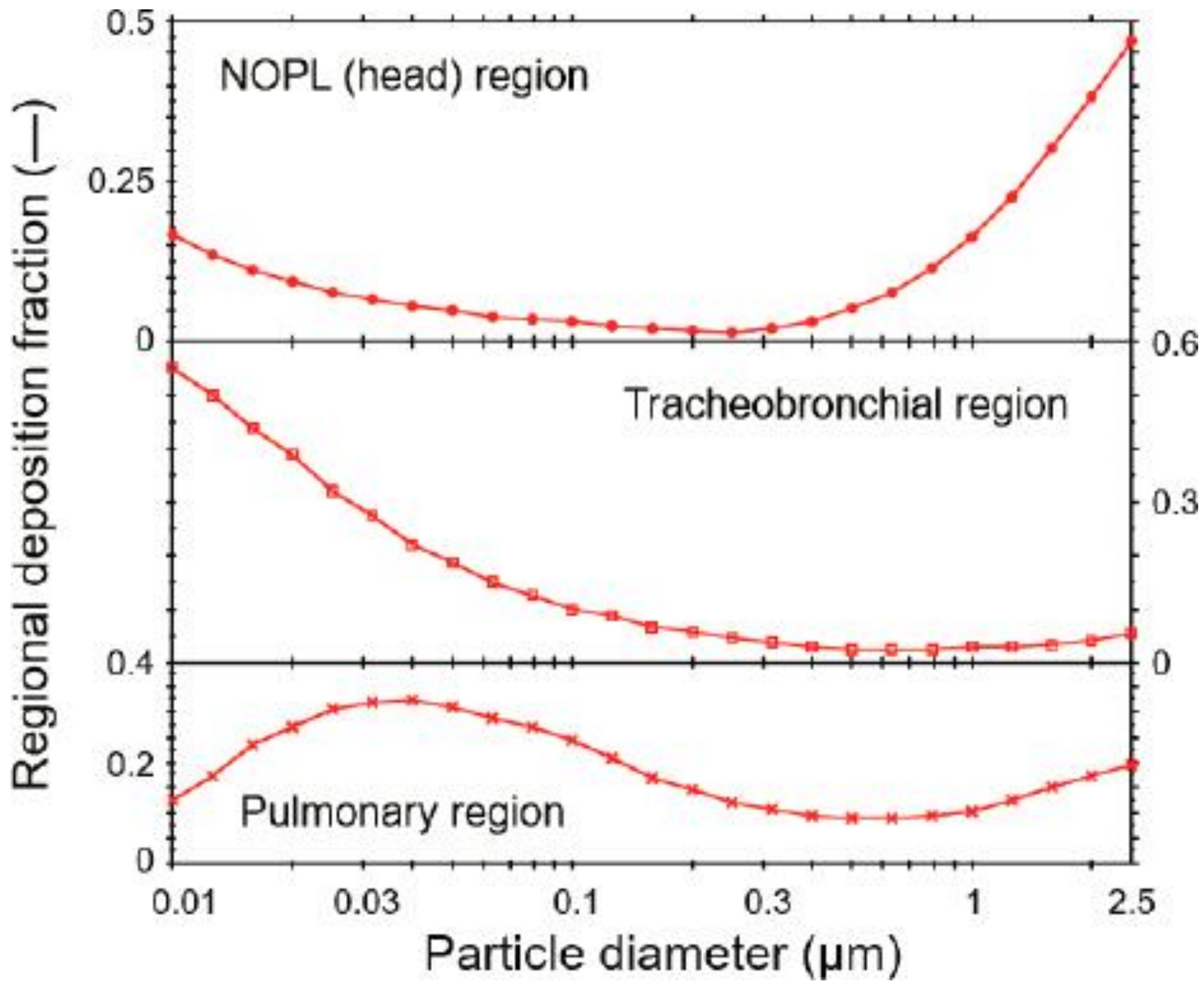


FIGURE 6-2 Size and spatial complexity of particle deposition in different regions of the lung. NOPL = nasal, oral, pharyngeal, and larynx. SOURCE: Nazaroff slide 7, adapted from Figure 1 in the cited publication.11 Comparisons of Calculated Respiratory Tract Deposition of Particles Based on the Proposed NCRP Model and the New ICRP66 Model. Yeh H-C, Cuddihy RG, Phalen RF, Chang IY. Aerosol Science and Technology 25(2):134–40. ©The American Association for Aerosol Research; http://www.tandfonline.com on behalf of The American Association for Aerosol Research, reprinted by permission of Taylor & Francis Ltd. National Academies of Sciences, Engineering, and Medicine. 2022. Indoor Exposure to Fine Particulate Matter and Practical Mitigation Approaches: Proceedings of a Workshop. Washington, DC: **The National Academies Press. https://doi.org/10.17226/26331.**

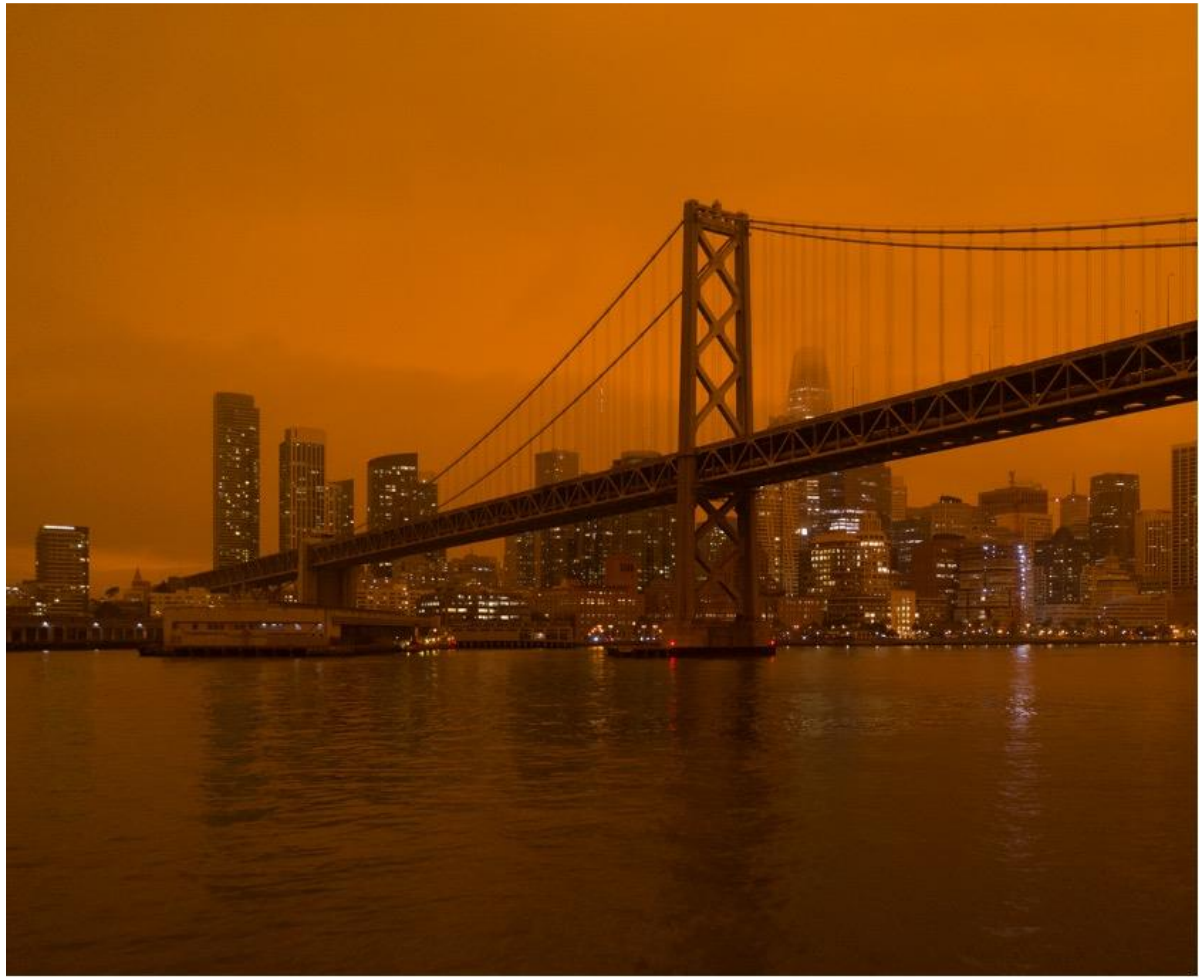
How to reduce indoor exposure to wildfire PM2.5

Best Practices

- Shelter indoors: buildings are the first line of protection from PM2.5 exposure
- Use a portable air cleaner (air purifier)
- If you have one, use your mechanical air ventilation with high efficiency filter

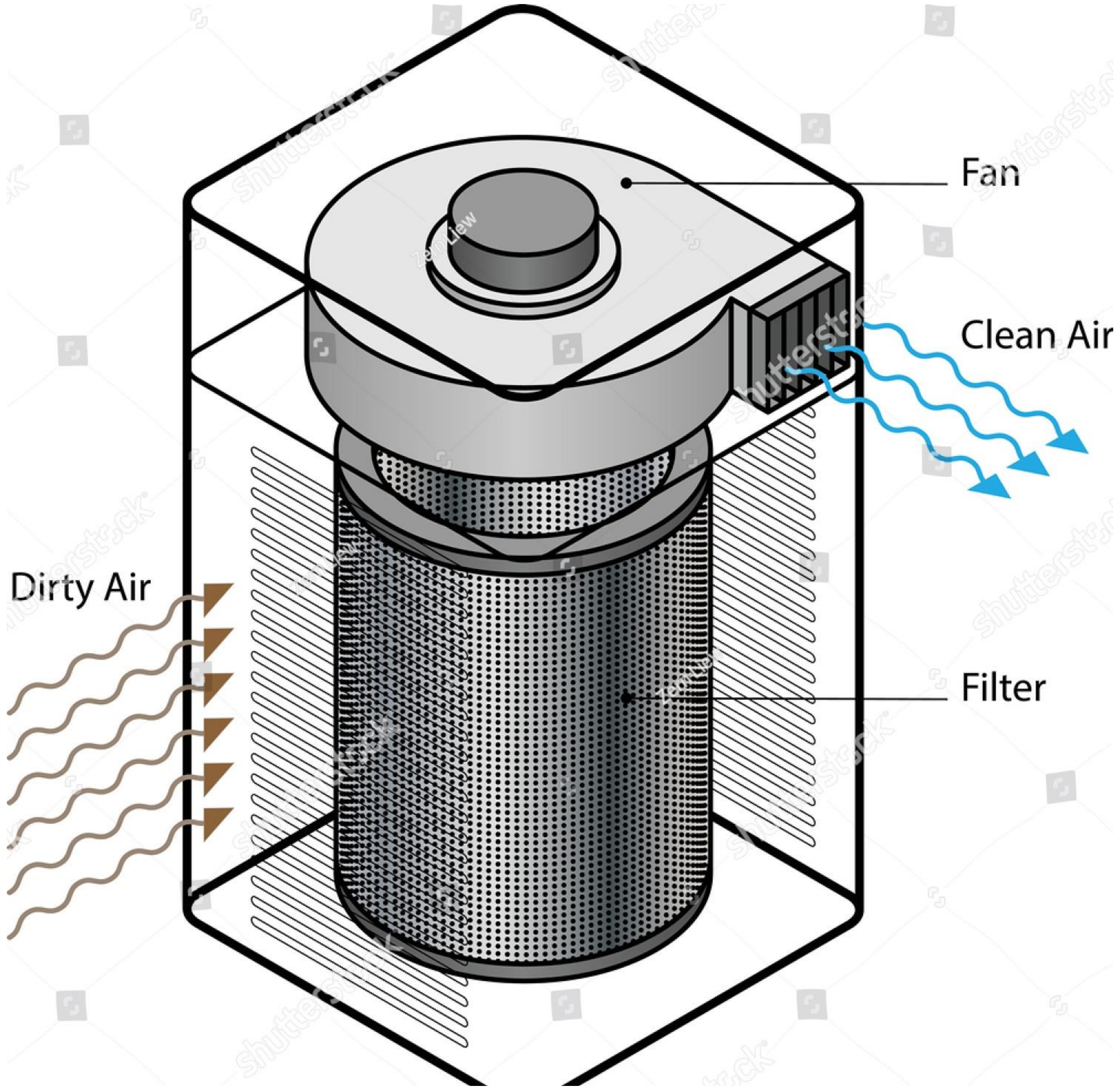
Barrier

- Heavily dependent on occupants' knowledge and behaviour

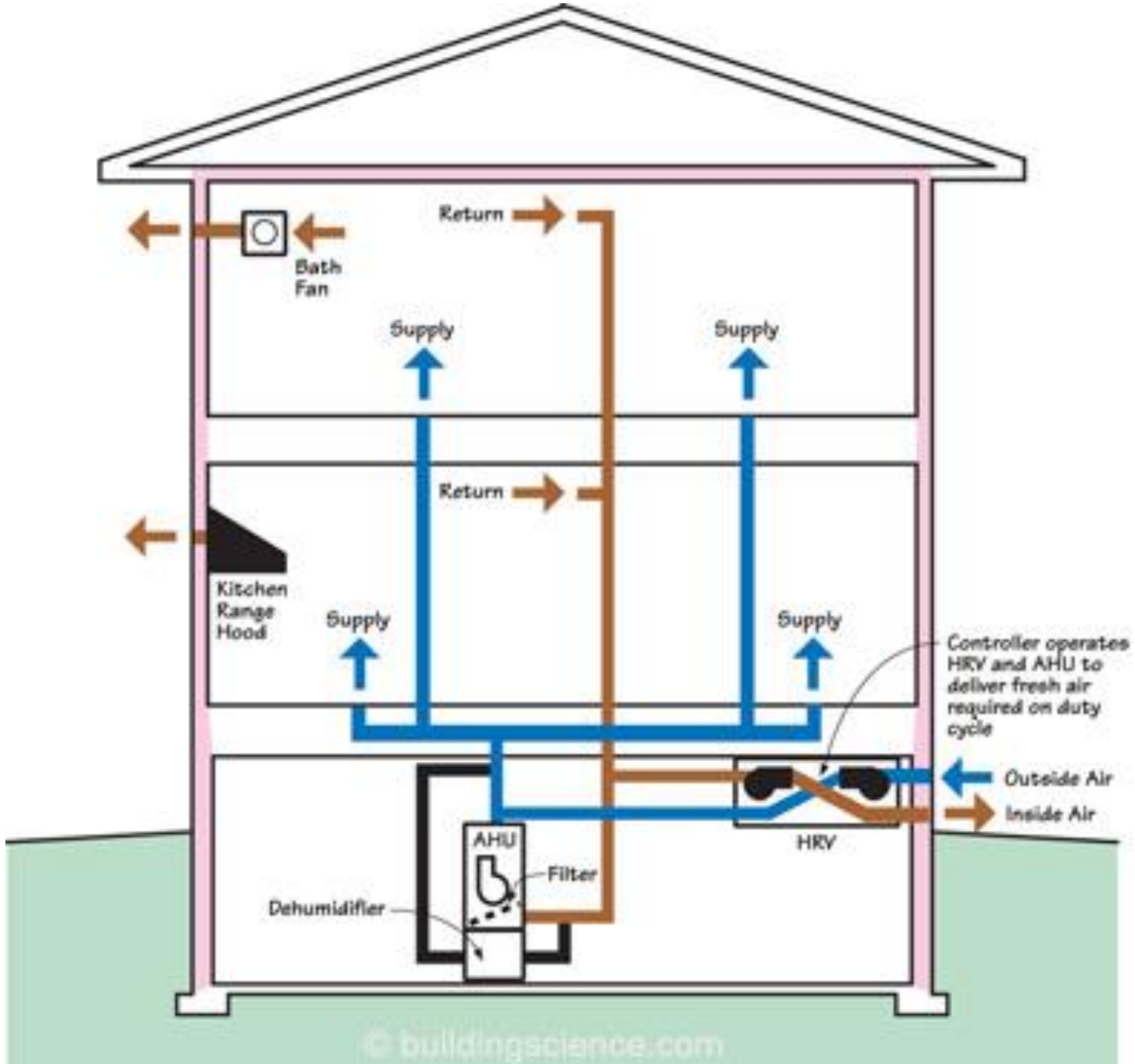


Smoke from the North Complex Fire settles over San Francisco.
Image credits: Christopher Michel from Flickr)

In short: what is an HVAC (~ VMC)



Air Purifier



Mechanical Air Ventilation

Overview

Objective

- Study households HVAC usage during wildfires
- Define HVAC best practices for reducing indoor exposure to wildfire PM2.5

Approach

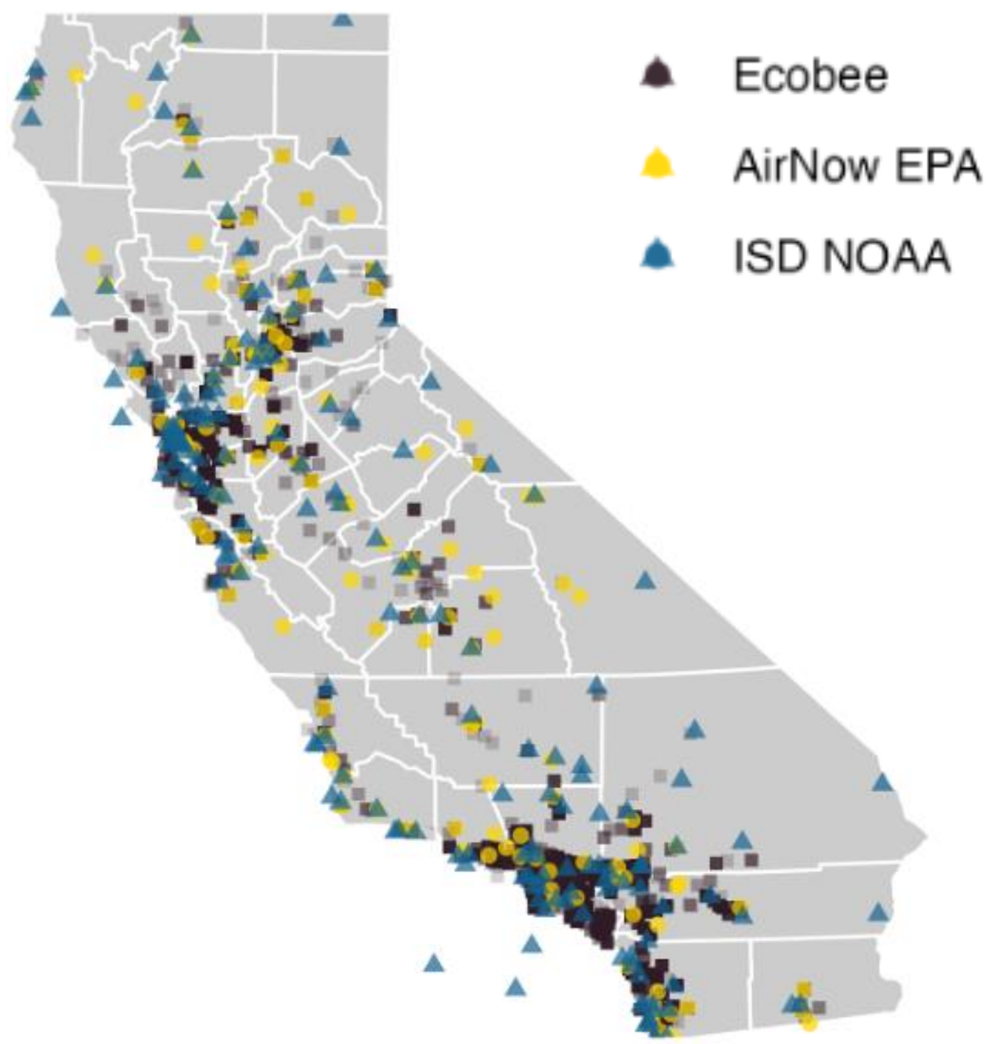
- Use smart-thermostat open data to analyse HVAC runtime during wildfire days
- Estimate the benefit of using a smart thermostat control logic to automatically run the HVAC during wildfire days
- Data from August to September 2020

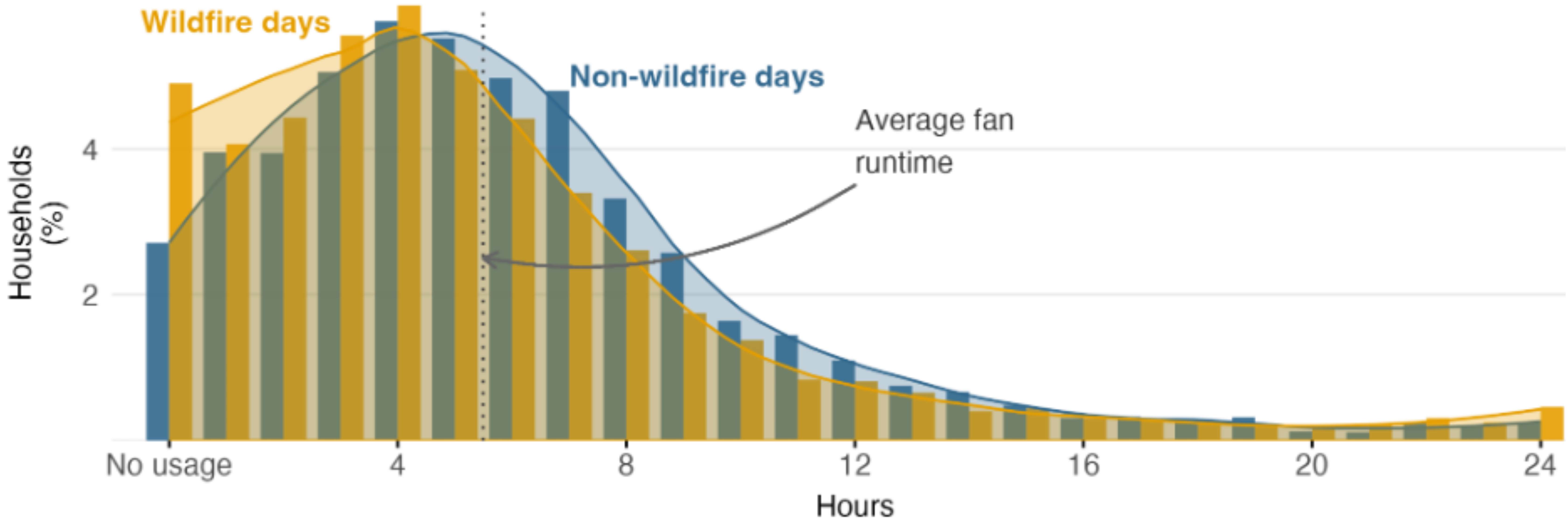
Hardware implementation for PM2.5 exposure reduction:
- An Ecobee smart-thermostat on the left
- Indoor PM2.5 monitor on the right

Fundings

- CITRIS and the Banatao Institute ($ 50k)
- Center for the Built Environment ($ 10k)
- EU Marie-Sklodowska Curie Action (https://cordis.europa.eu/project/id/844526)

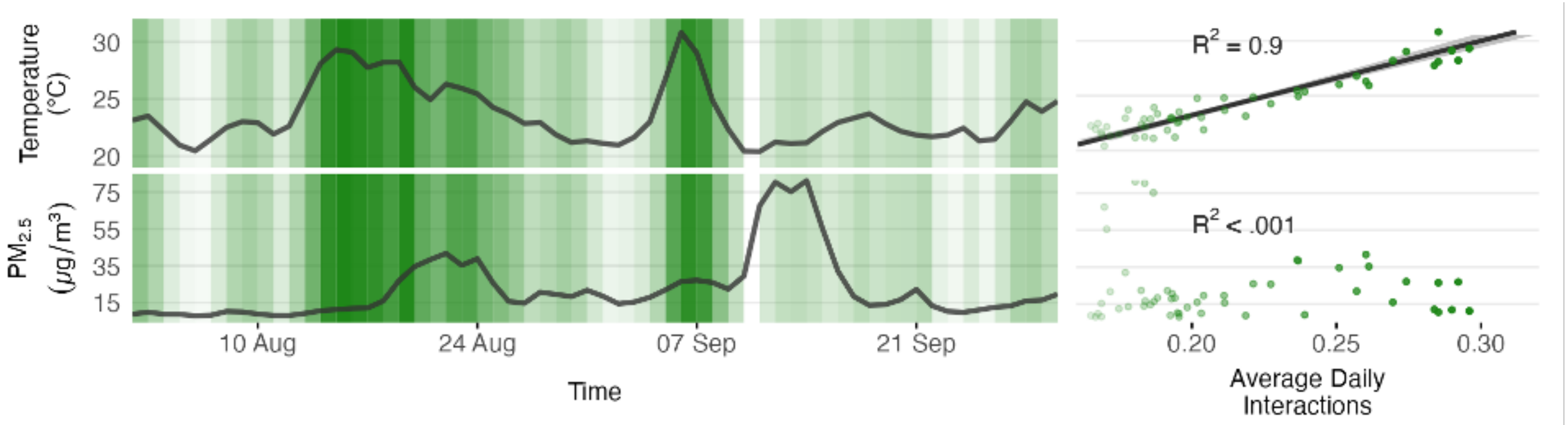
Smart thermostat data: fan runtime





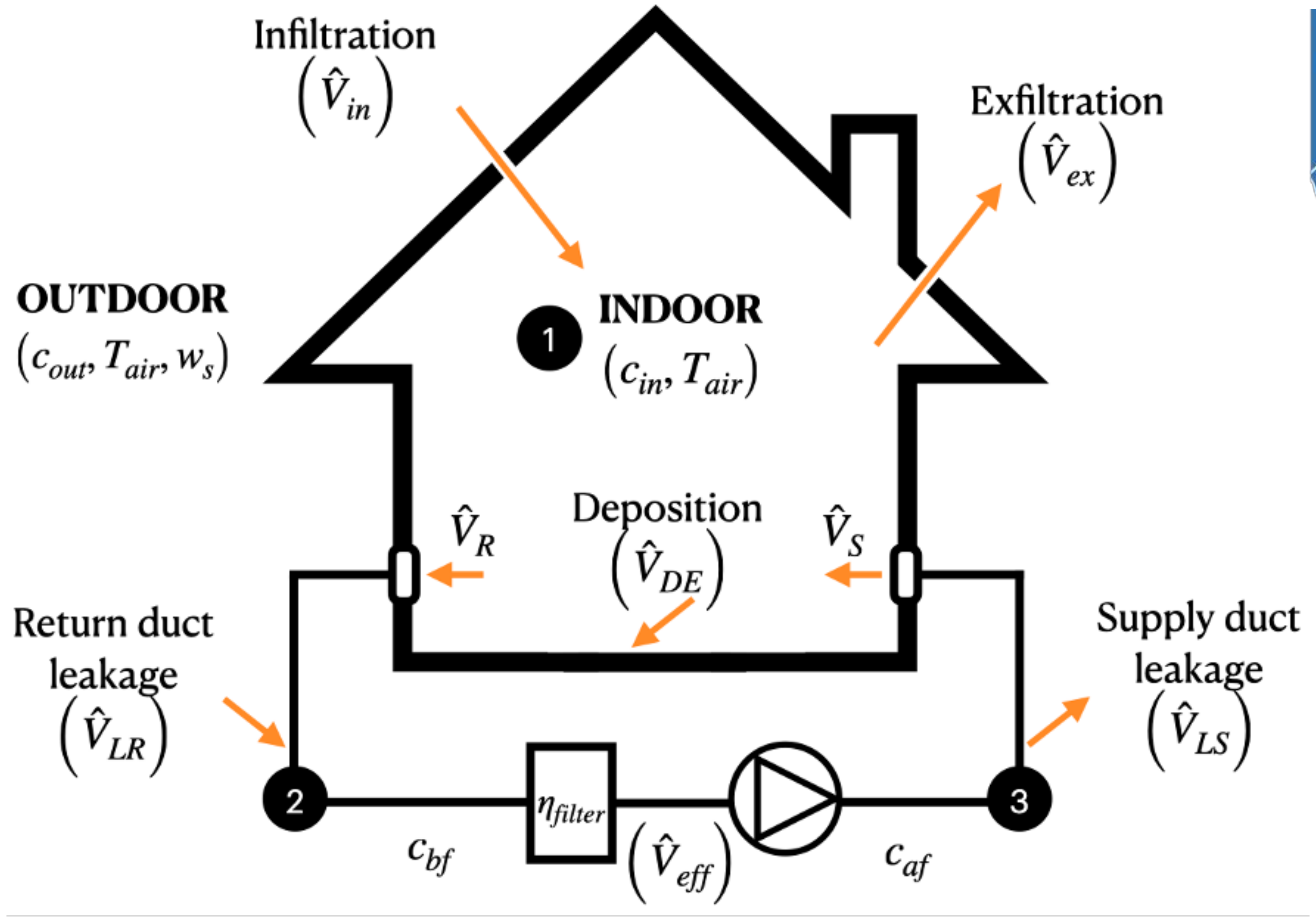
- 4,950 Ecobee smart thermostat, 159 NOAA monitors and 132 EPA stations
- Daily median fan runtime: 5 hr 30 min
  - Wildfire day: 5 hr 18 min
  - Non-wildfire day: 5 hr 32 min
- ~5% of houses do not use the HVAC at all

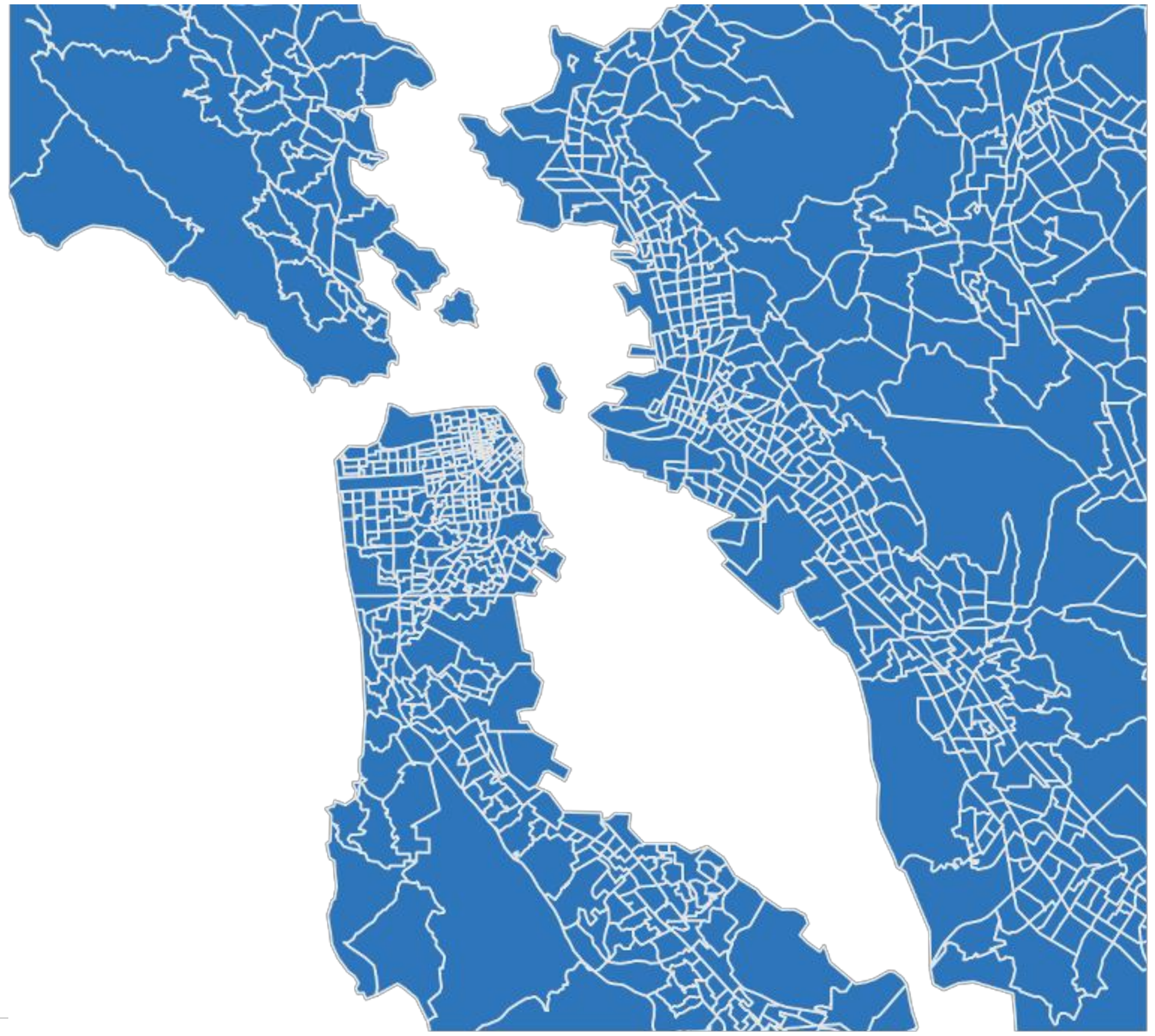
Smart thermostat data: occupants' behaviour



- Manual interactions with the smart thermostats nearly double with a 10°C (18°F) increase in average outdoor temperature
- There is no relationship between interactions with thermostat and outdoor PM2.5
- <u>Households did not rely on HVAC during wildfire days to improve indoor air quality</u>

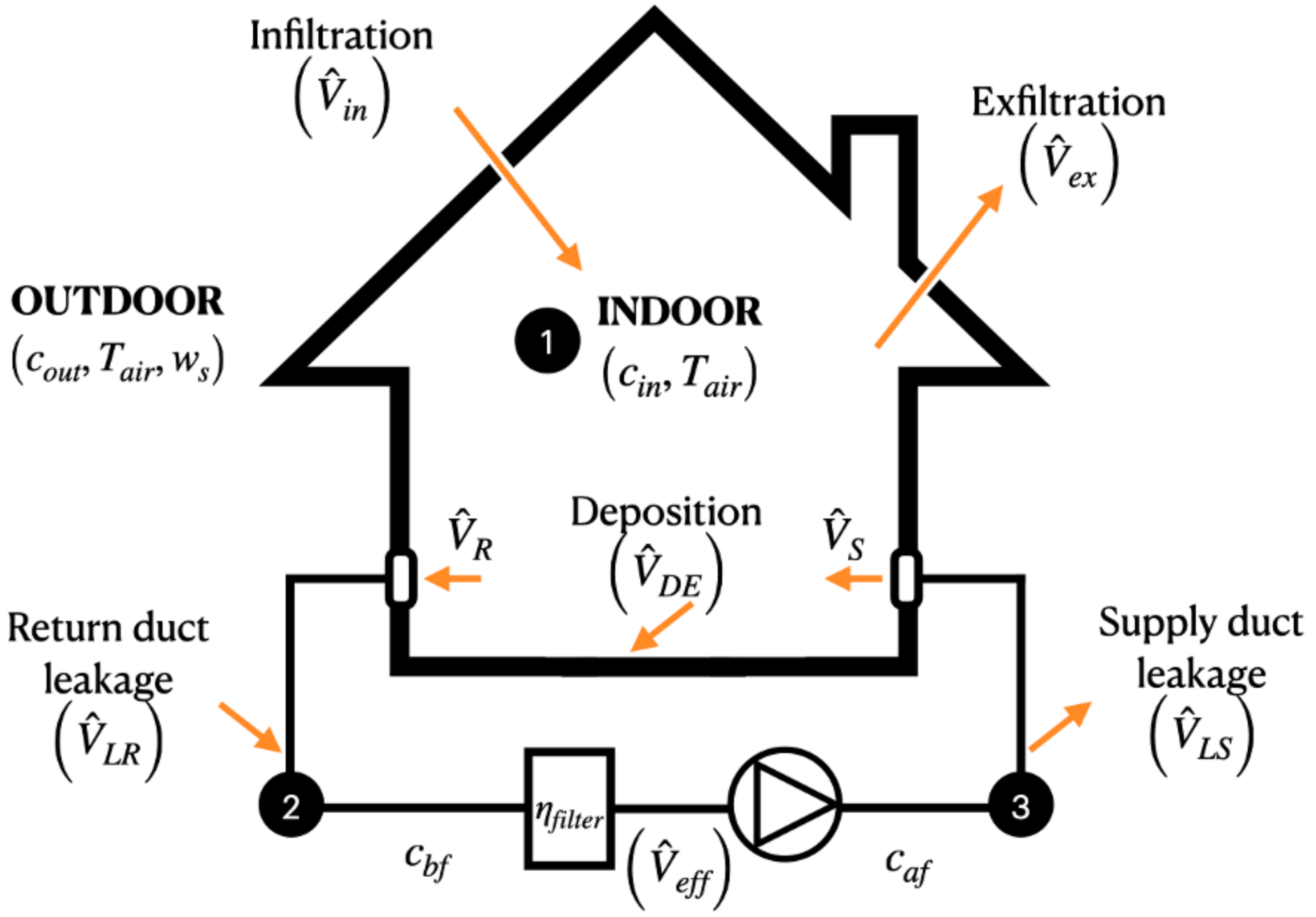
Model for estimating indoor PM2.5 concentration





- We built a synthetic dataset of about 9,000 Californian homes using
  - The American Community Survey
  - ASHRAE Climate Zones

Model for estimating indoor PM2.5 concentration



physical airflows terms $(\hat{V}\ [\frac{m^3}{s}])$

$$c_{in,t+\Delta t} = c_{in,t} + \frac{1}{V}\left(\left[\hat{V}_{in,t} \cdot c_{out,t} \cdot k_P\right] - \left[\hat{V}_{ex,t} \cdot c_{in,t}\right] - \left[\hat{V}_{DE,t} \cdot c_{in,t}\right] - i \cdot \left[\hat{V}_{eff} \cdot (1-\eta_{LR}) \cdot c_{in,t}\right] + i \cdot \left[\hat{V}_{eff} \cdot (1-\eta_{LS}) \cdot c_{af,t}\right]\right) \cdot \Delta t$$

Model for estimating indoor PM2.5 concentration

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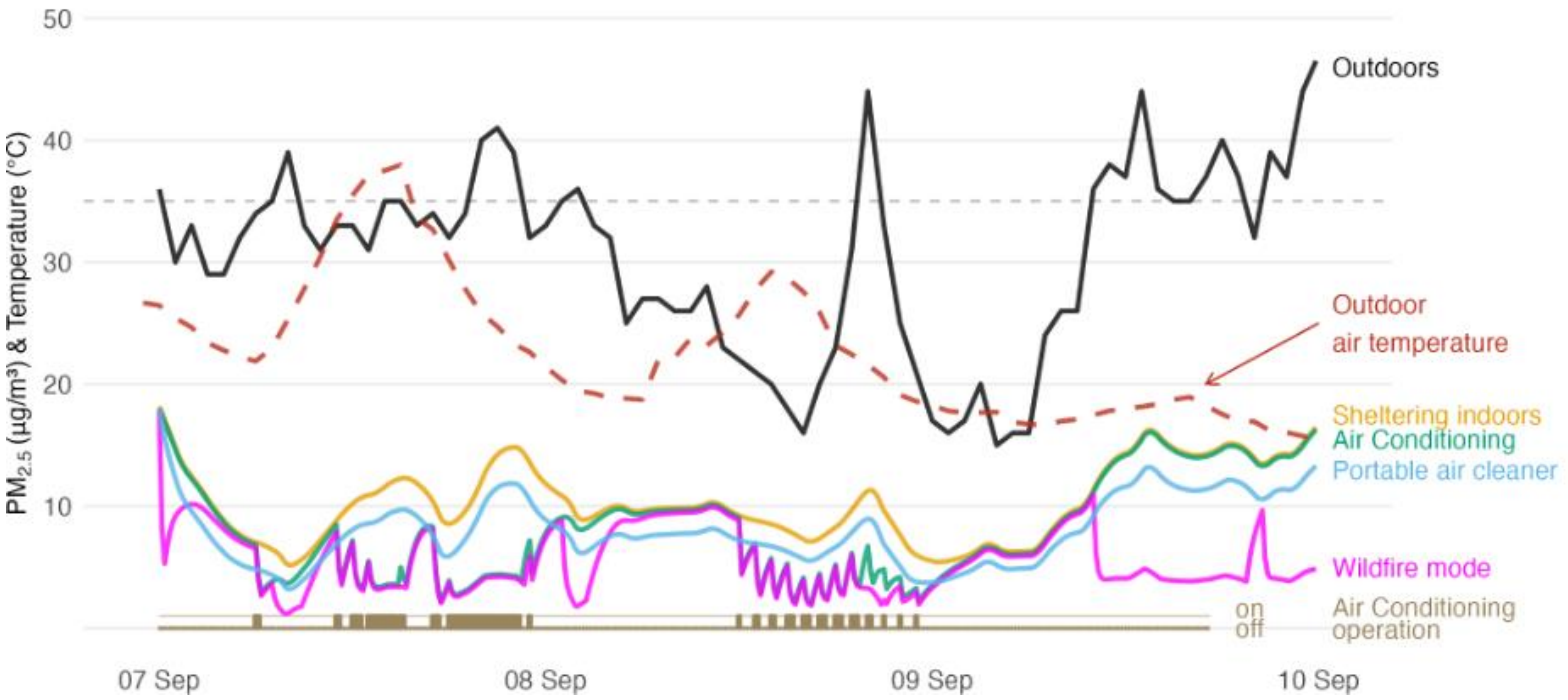
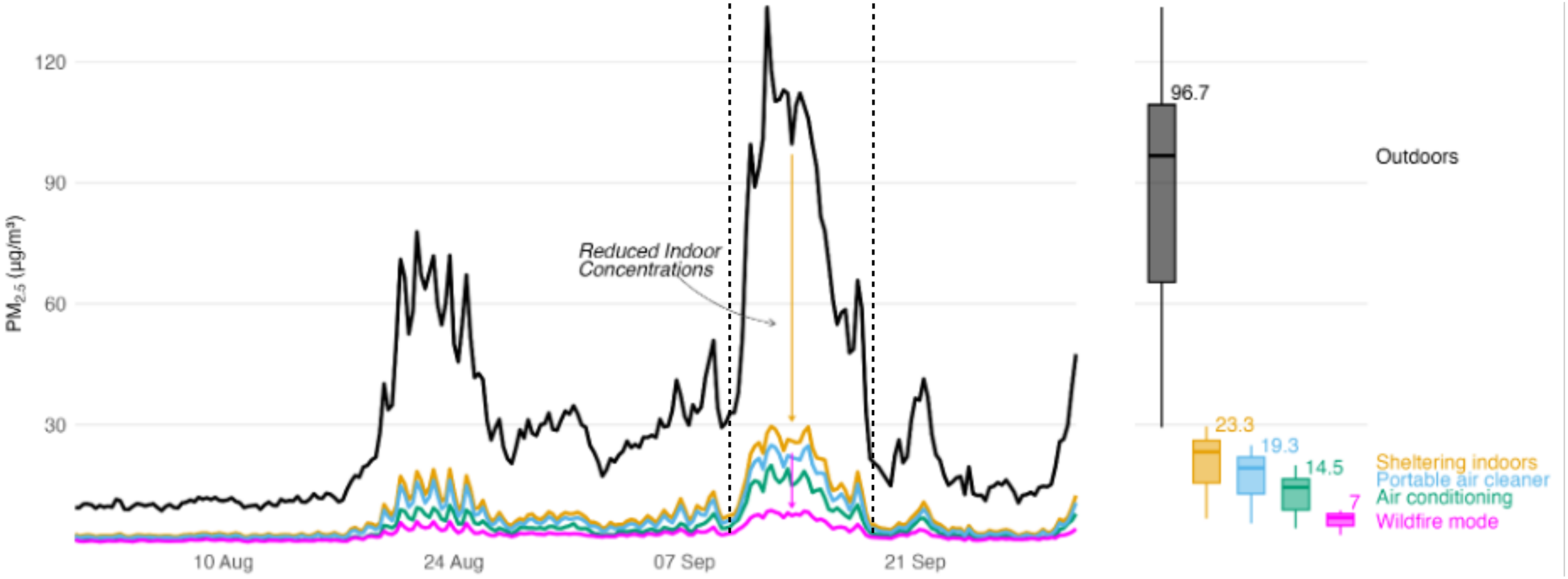


- We modelled HVAC PM2.5 removal efficiency
- Residential HVAC is as effective as about 4 air purifiers on average

Potential benefit of using HVAC



Sensitivity Analysis – Monte Carlo

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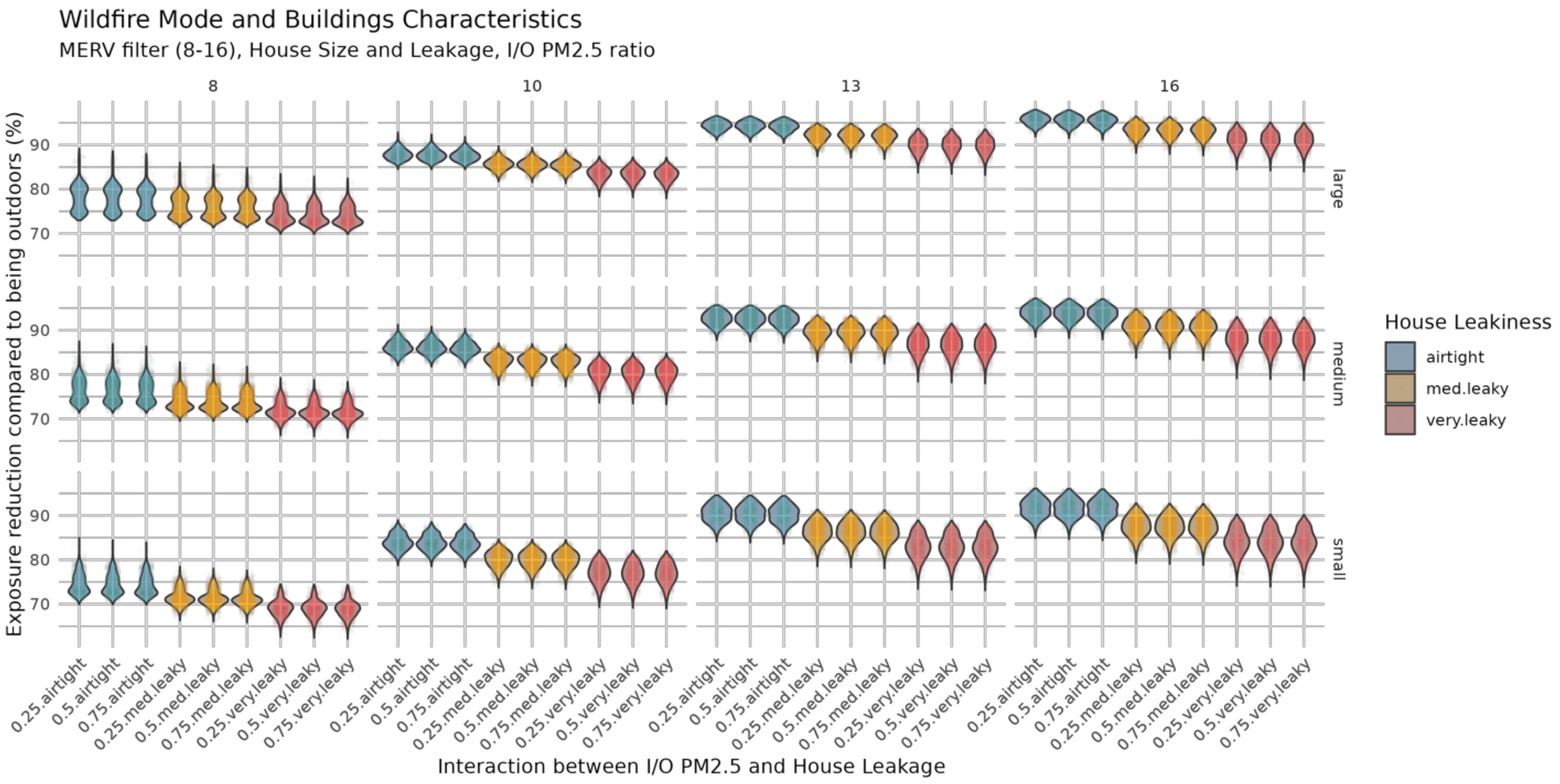
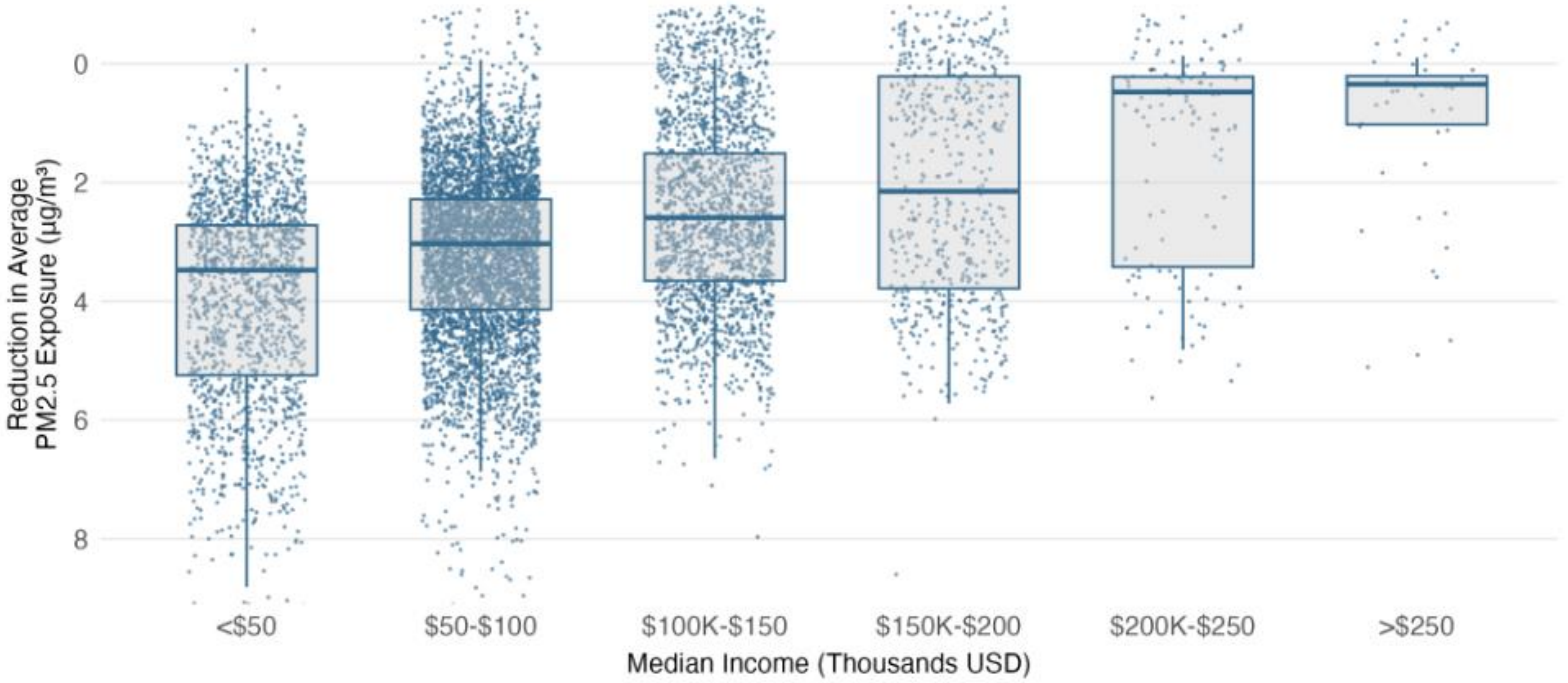
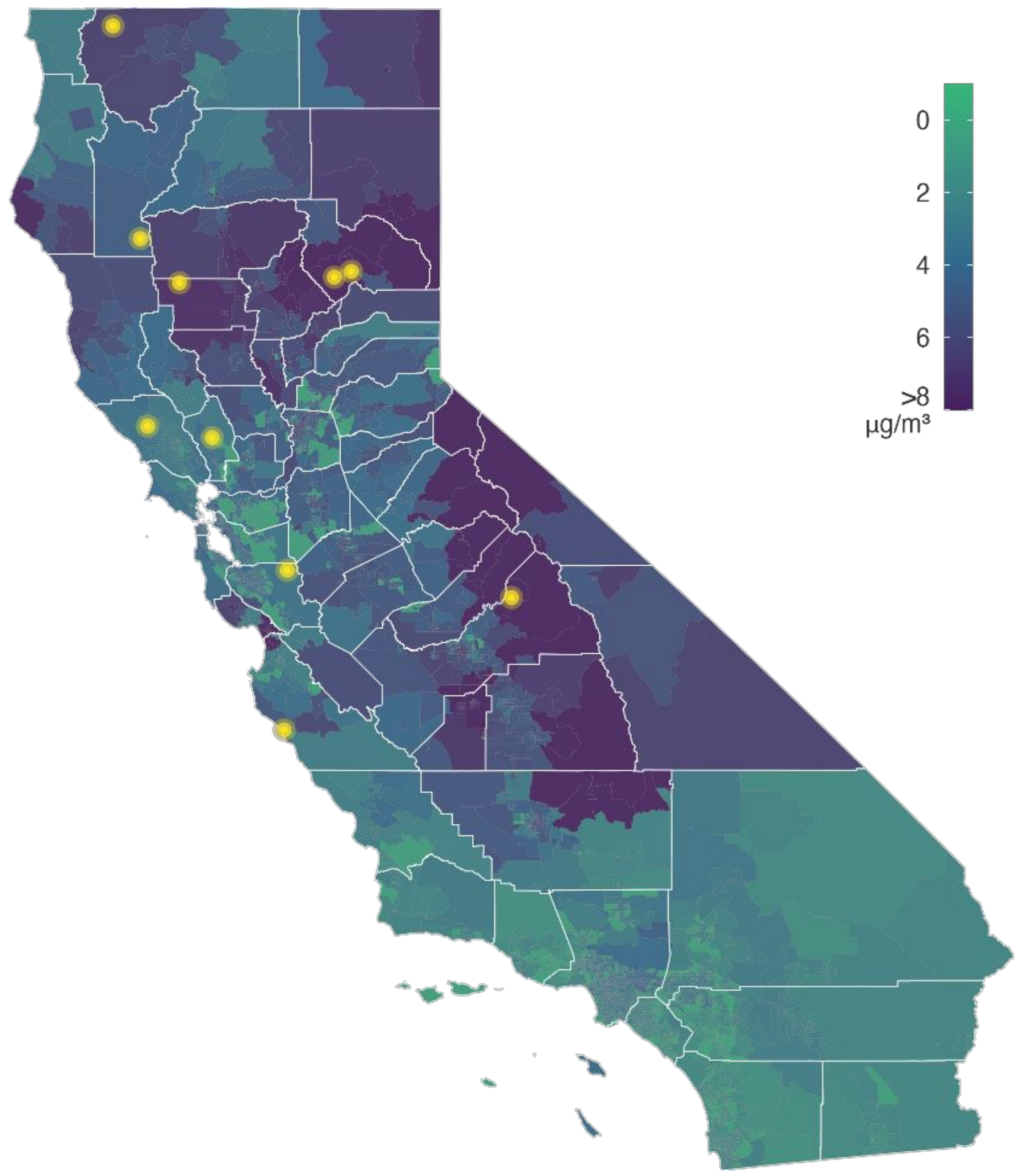
Additional 187.000 Simulations

- House floor area
- Effective leakage area of the building envelope
- Initial indoor/outdoor (I/O) PM2.5 ratio (0.25, 0.50, 0.75)
- Central air system filter efficiency (MERV 8, 10, 13, 16)
- Central air system runtime (sampled using Sobol sequence)



Health Impact Analysis





- The wildfire mode control strategy reduced mortality risk by 76%, with an estimated $29M in monetized benefits ($5 per month per household)
- The greatest potential to reduce the health impact of PM2.5 exposure lies in lower-income areas

EPA's BenMAP-CE software

J.D. Sacks, J.M. Lloyd, Y. Zhu, J. Anderton, C.J. Jang, B. Hubbell, N. Fann, The Environmental Benefits Mapping and Analysis Program–Community Edition (BenMAP–CE): A tool to estimate the health and economic benefits of reducing air pollution, Environ. Model. Softw. 104 (2018) 118–129.

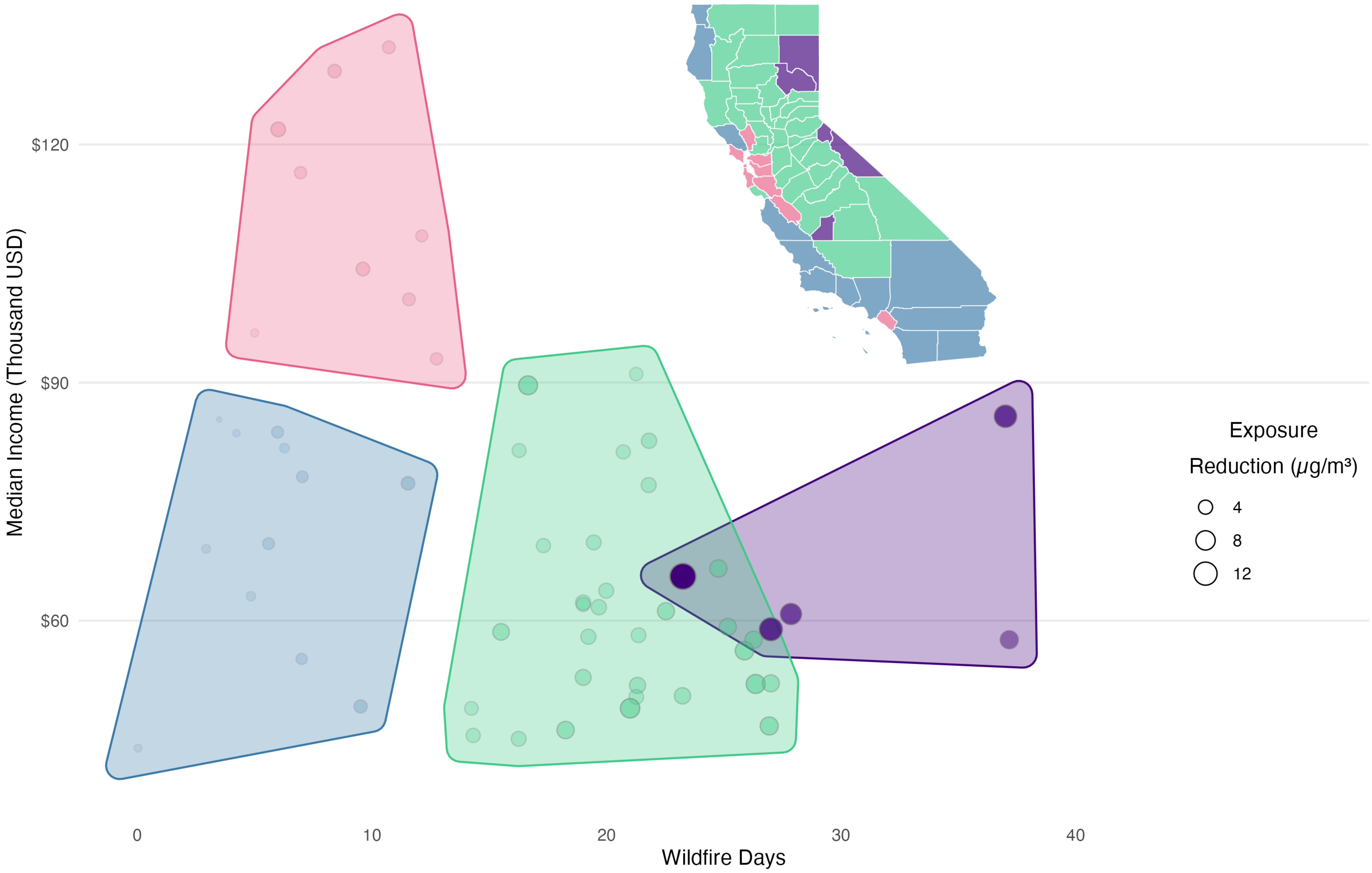
concentration-response effect estimate

$$\Delta y = \left(1 - e^{-\beta \Delta_{PM}}\right) \cdot y_o \cdot a \cdot P$$

change in the incidence rate from the baseline

Cluster Analysis

- ~8 ug/m^3 reduction near wildfires
- ~5 ug/m^3 reduction in the Central Valley (21% of the CA population)
- ~3 ug/m^3 reduction in the SF Bay Area
- ~3 ug/m^3 reduction in LA despite the distance from wildfires (urban air pollution?)



Takeaways

- Sheltering indoors reduces exposure to outdoor PM2.5 levels by approximately 75%
- Air conditioning reduces exposure to particulate matter by 40%, achieving indoor PM2.5 levels that are 87% lower than those outdoors
- Wildfire mode reduces exposure to particulate matter by 54% or 90% compared to outdoor PM2.5 levels
- Central air systems can match the filtration efficiency of 4 portable air purifiers in an average California home
- Using wildfire mode increases system runtime by 46%, adding about $5 to monthly energy bills per California household

Dallo, Federico, et al. "Using smart thermostats to reduce indoor exposure to wildfire fine particulate matter (PM2. 5)." Indoor Environments (2025): 100088.

https://github.com/CenterForTheBuiltEnvironment/tstat-wildfire

Smart thermostat control logic

Inputs

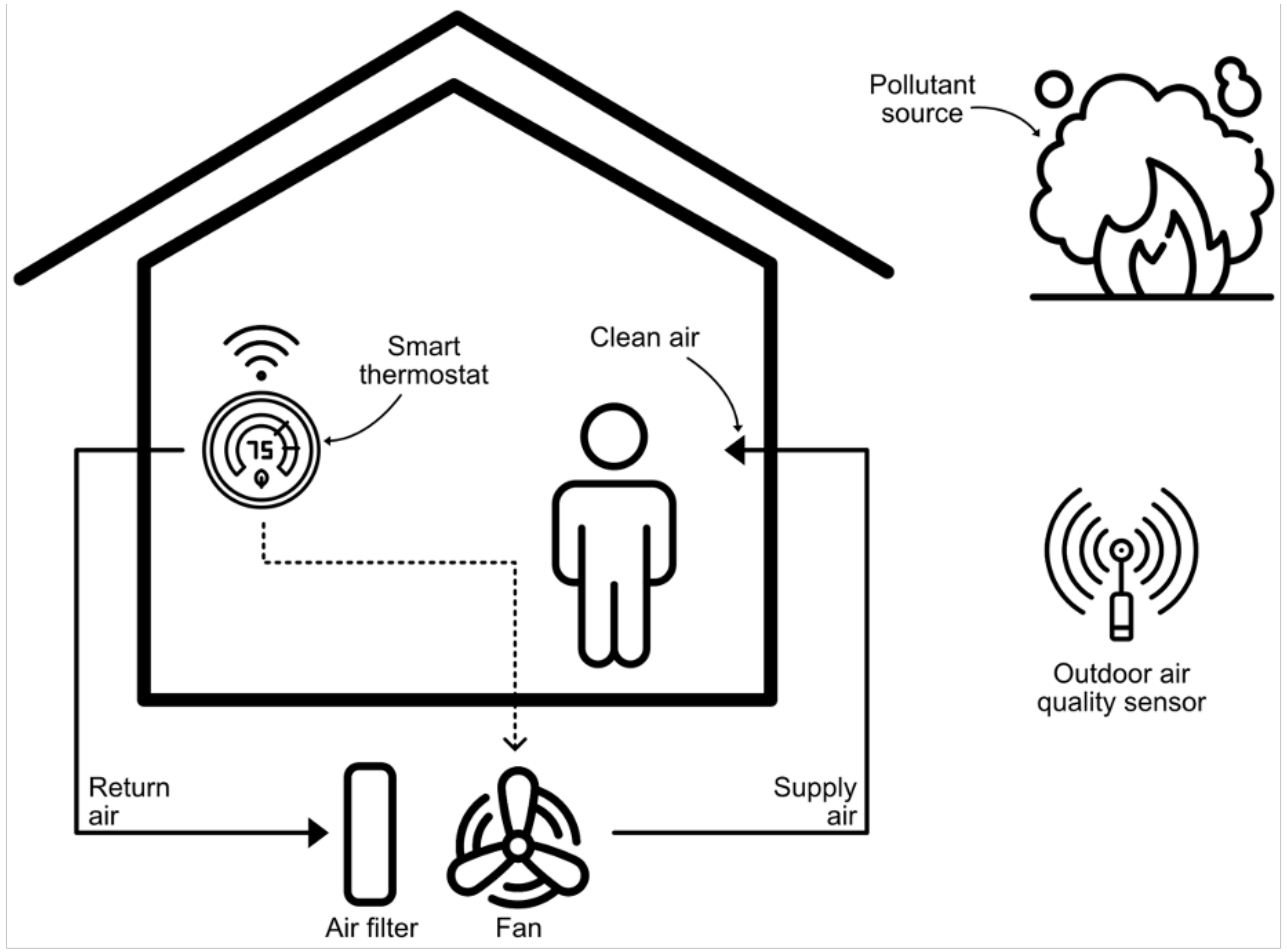
- Occupancy
- Outdoor PM2.5

Optional

- House characteristics
  - Floor Area
  - Age
- Indoor PM2.5

Output

- Optimized HVAC fan runtime



Dallo et al. In Prep.

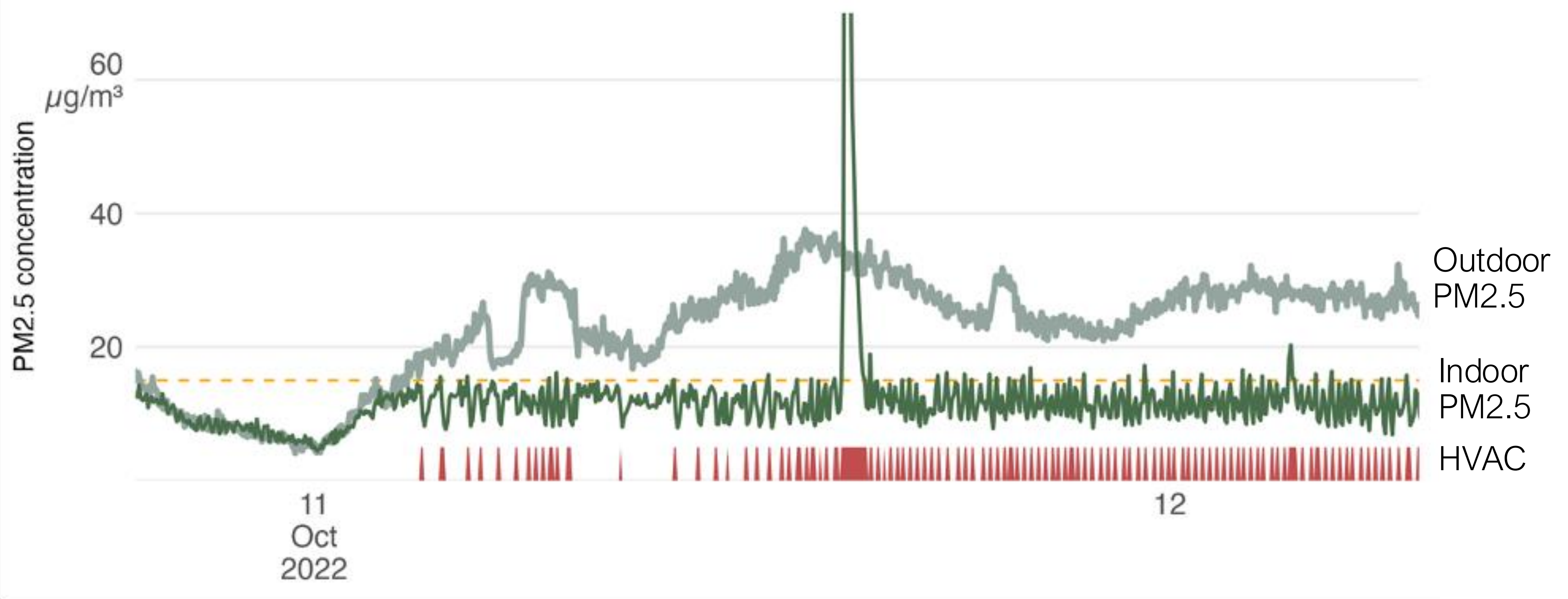
Field Study

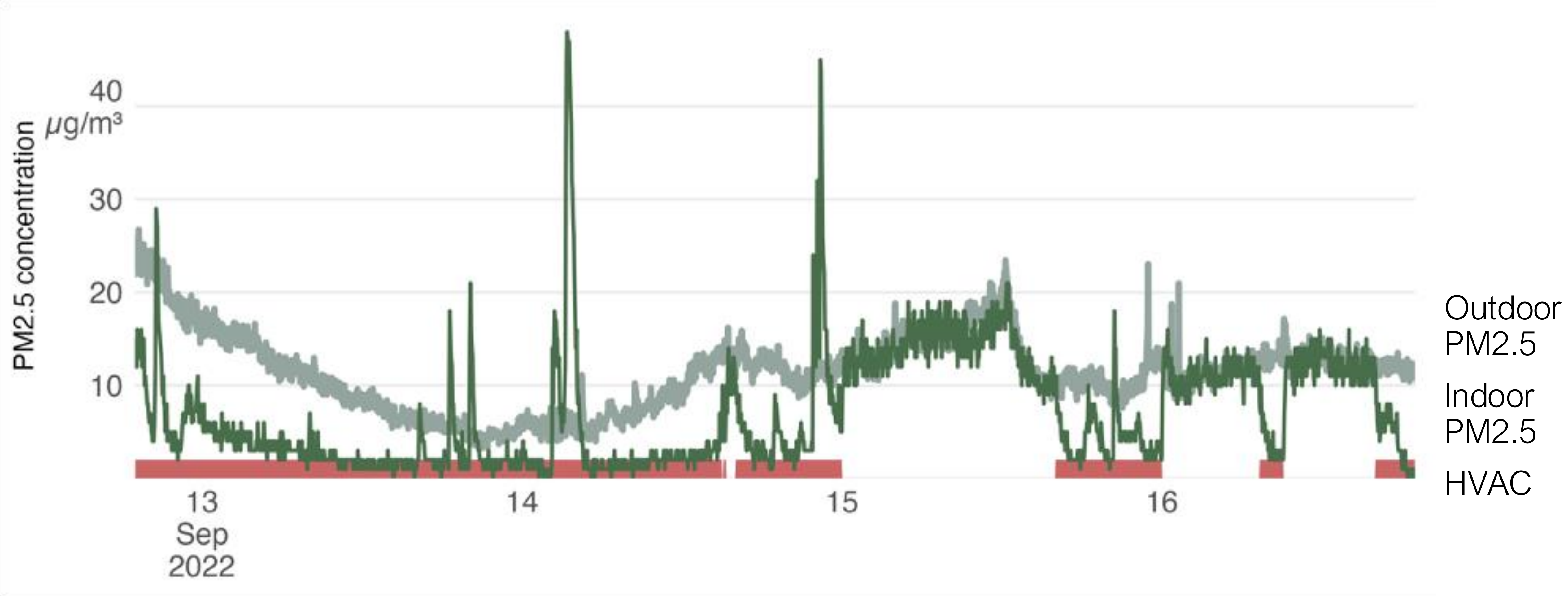
Small Apartment

- 48% of indoor PM2.5 reduction in the period considered

Detached House

- 30% of indoor PM2.5 reduction in the period considered





Dallo et al. In Prep.

Conclusions

Behaviour

- Households were not using the HVAC system to reduce indoor PM2.5 concentration

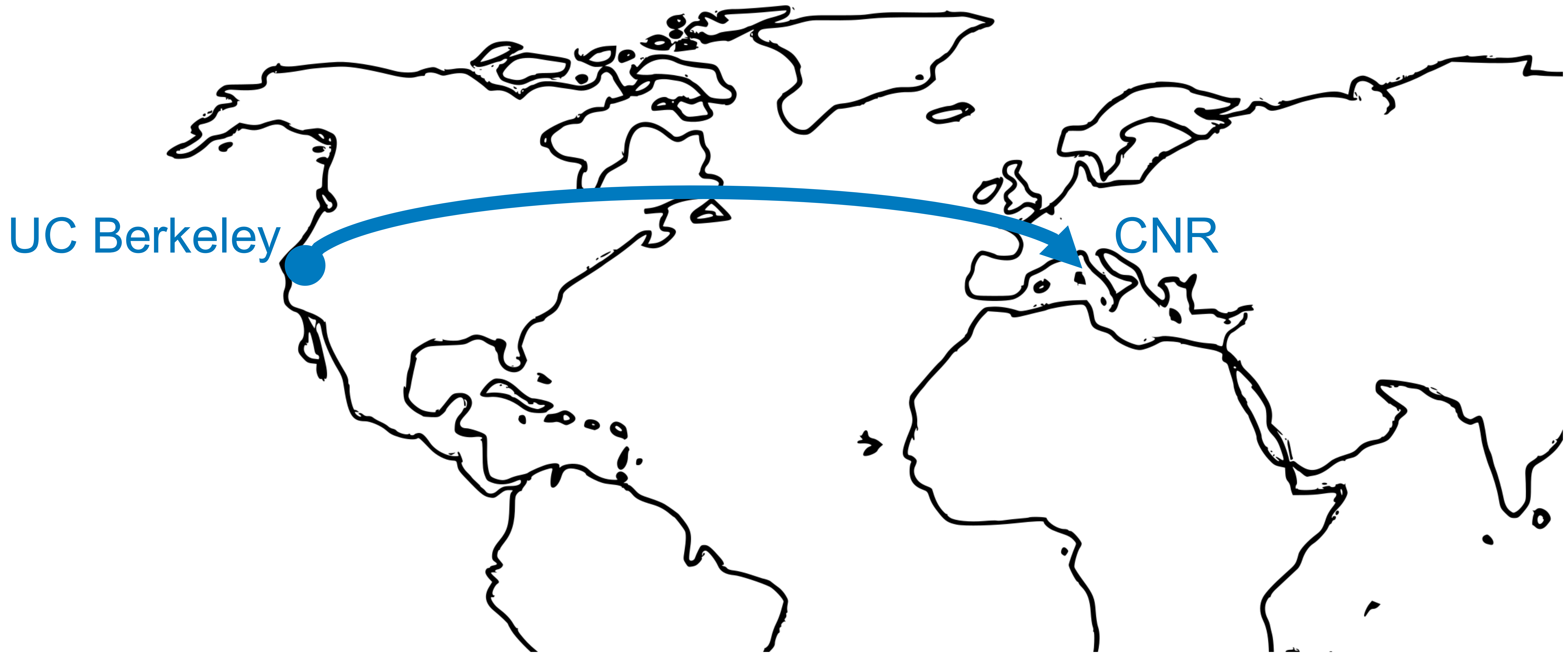
HVAC Performances

- Equivalent to about 4 portable air cleaners
- Up to 50% reduction in indoor exposure to PM2.5

Smart Thermostat Control Logic

- Outdoor PM2.5 measurements can be used to automatic control of HVAC fan to reduce indoor PM2.5 exposure

A view of the San Francisco Bay on Nov. 14, 2018 (left) and three weeks prior.
Image from Berkeley Lab, Kelly J. Owen.



healthRiskADAPT project → Tools

11-2024 → 11-2028

Resilience to Climate Stressors

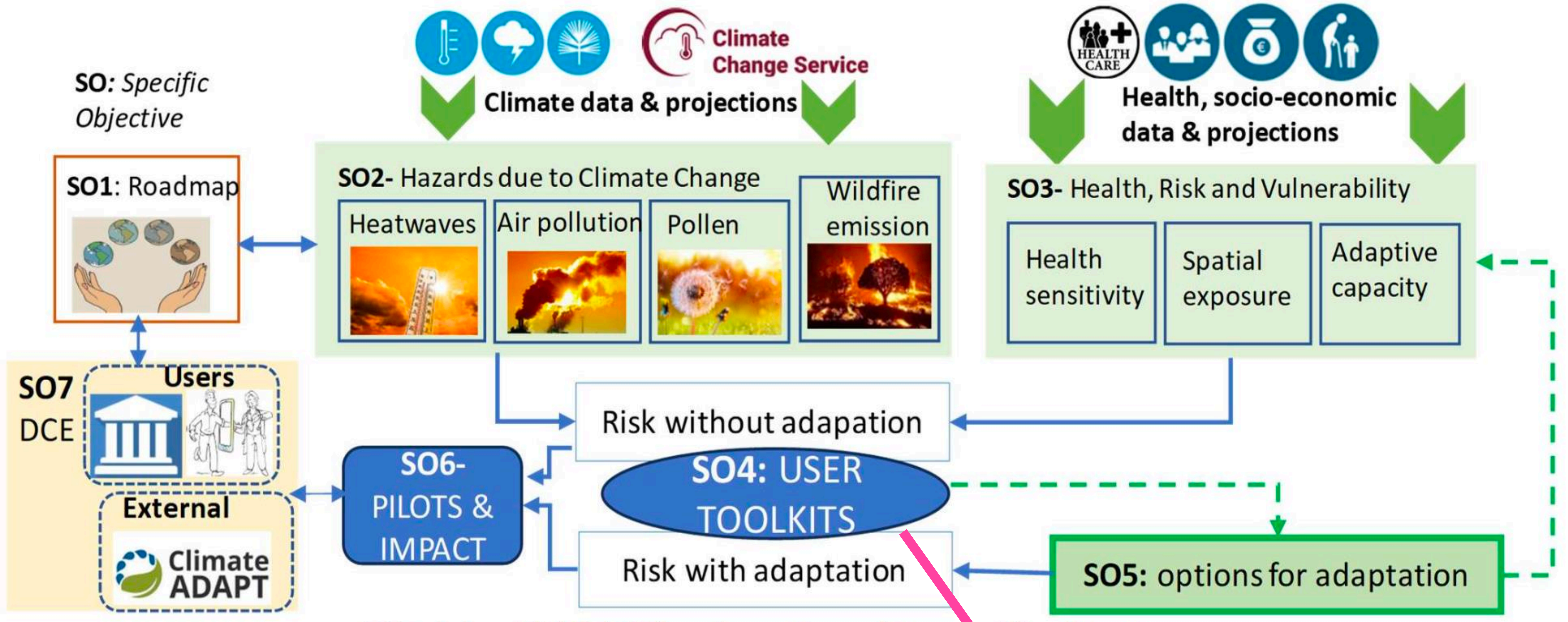


Fig. 1.1 *riskADAPT's schematic and its specific objectives*

CNR

Resilience to Climate Stressors

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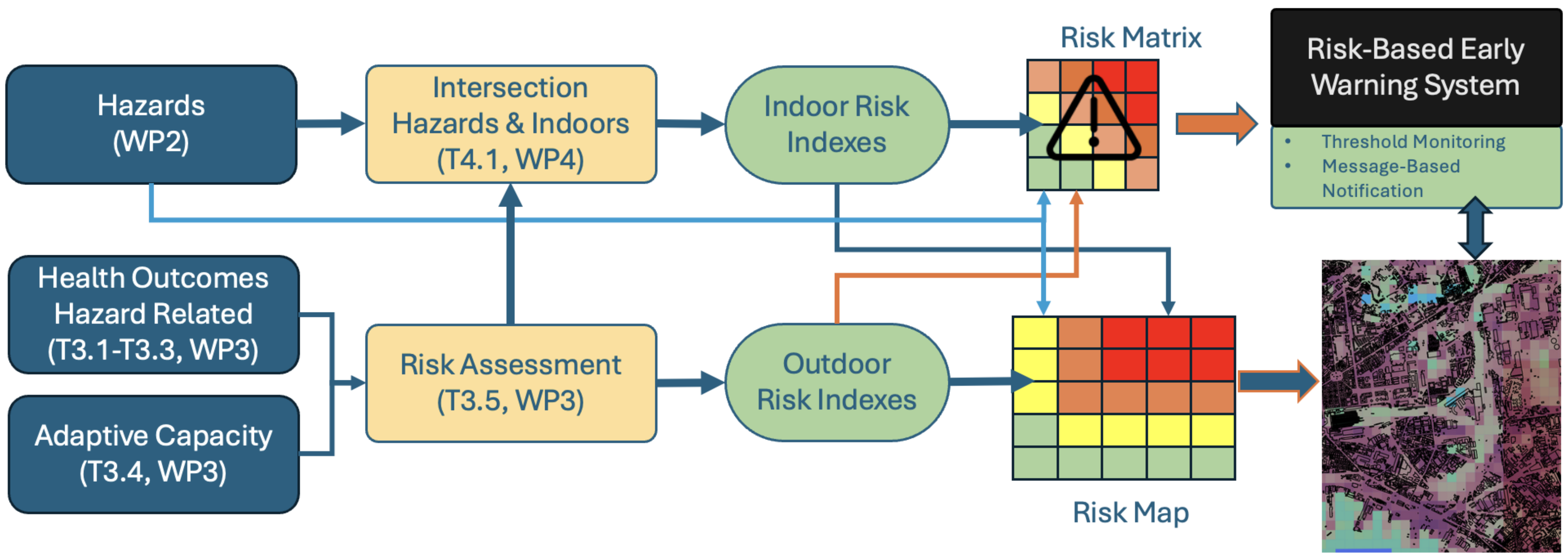
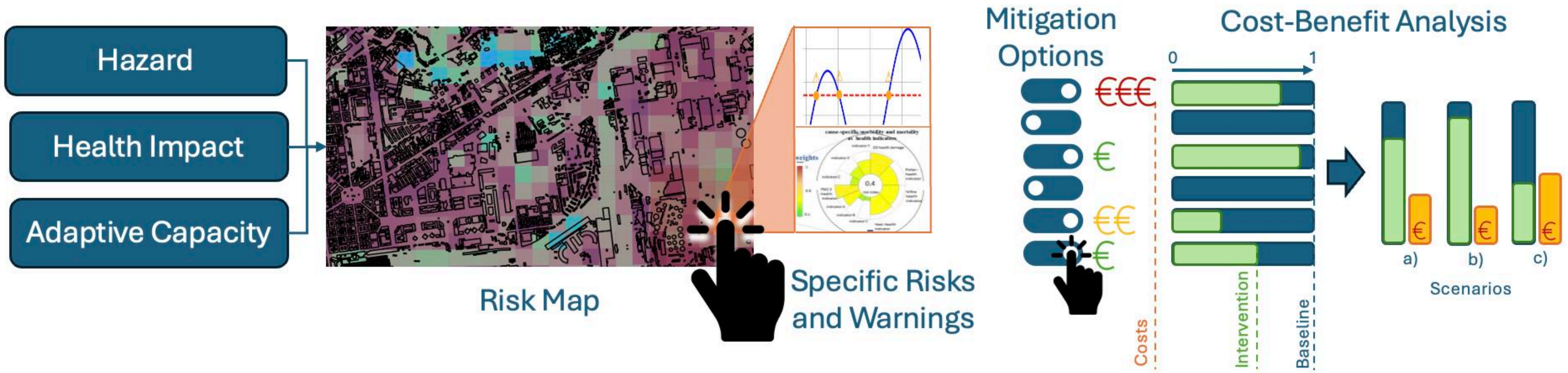
ISP Istituto di Scienze Polari
Consiglio Nazionale delle Ricerche

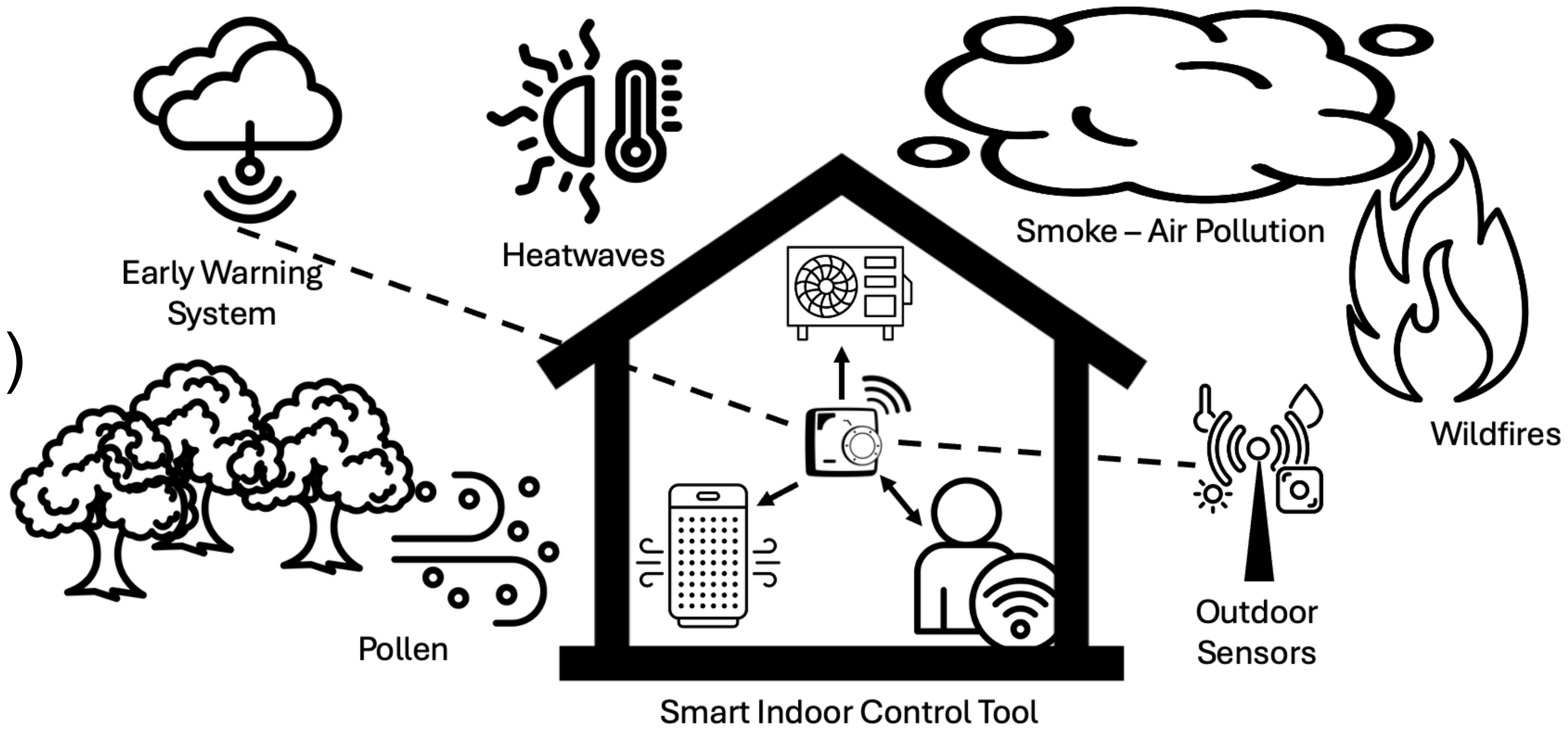


Oslo
Bern
Lyon
Naples*



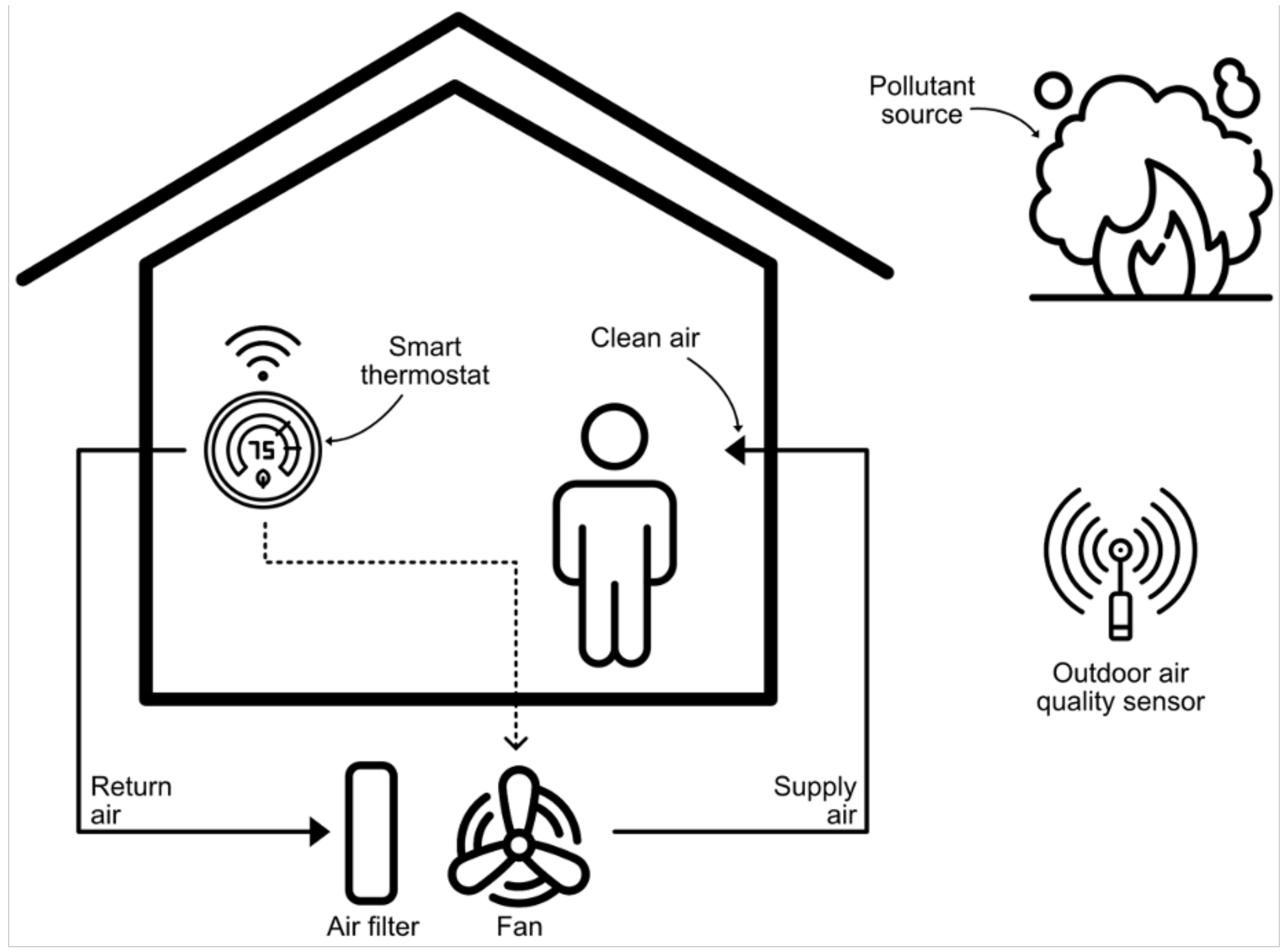
Can we disregard the indoor environments?

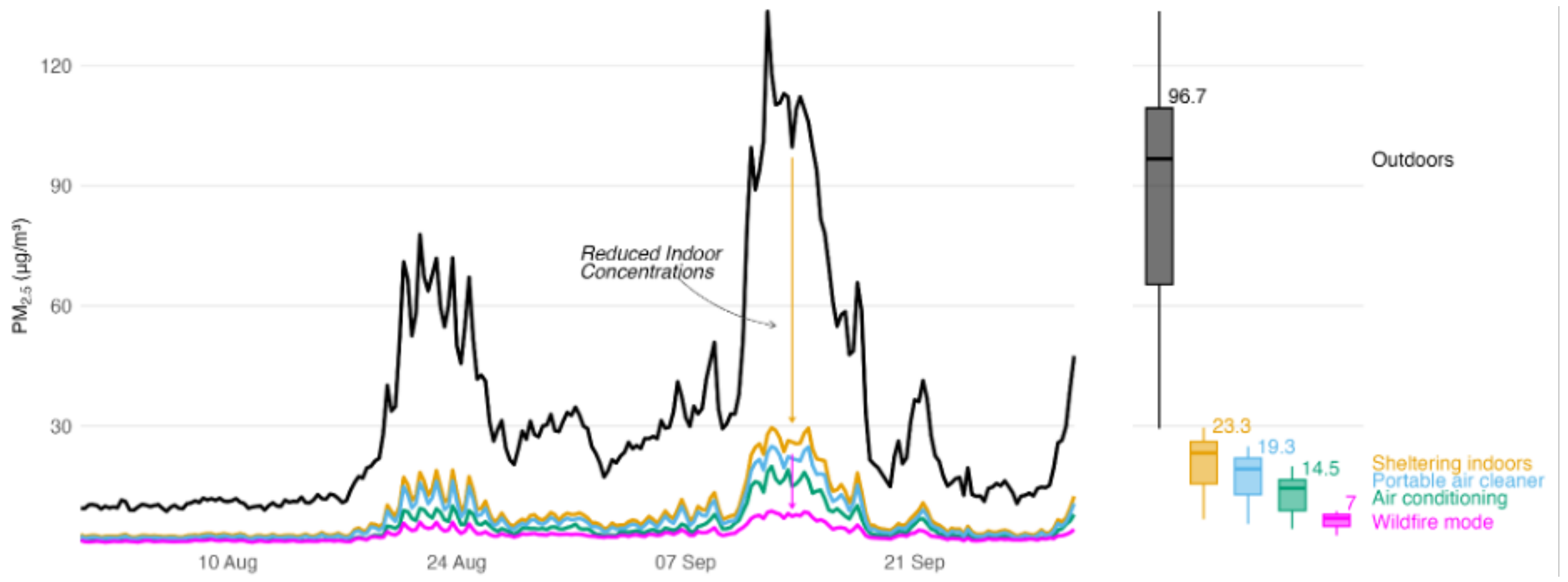
- **90% of exposure to air pollution happens indoors**
  - Infiltration of outdoor air pollution
  - Indoor activities (e.g., cooking, cleaning, … smoking)
- The indoor environment is VERY complex
  - Building type (age, size, insulation, equipement installed, …)
  - Occupants (number, age, behaviour, …)
  - Location (climate, sources, built environments, …)
  - Indoor and outdoor sources of air pollution
  - Human Behaviour and Gender Dimension
  - …



Exposure to Air Pollution: Largely Unknown, with Health Impacts Accounting for Only 10% of time spent outdoors → We need tools to measure indoor spaces

Existing Methods…





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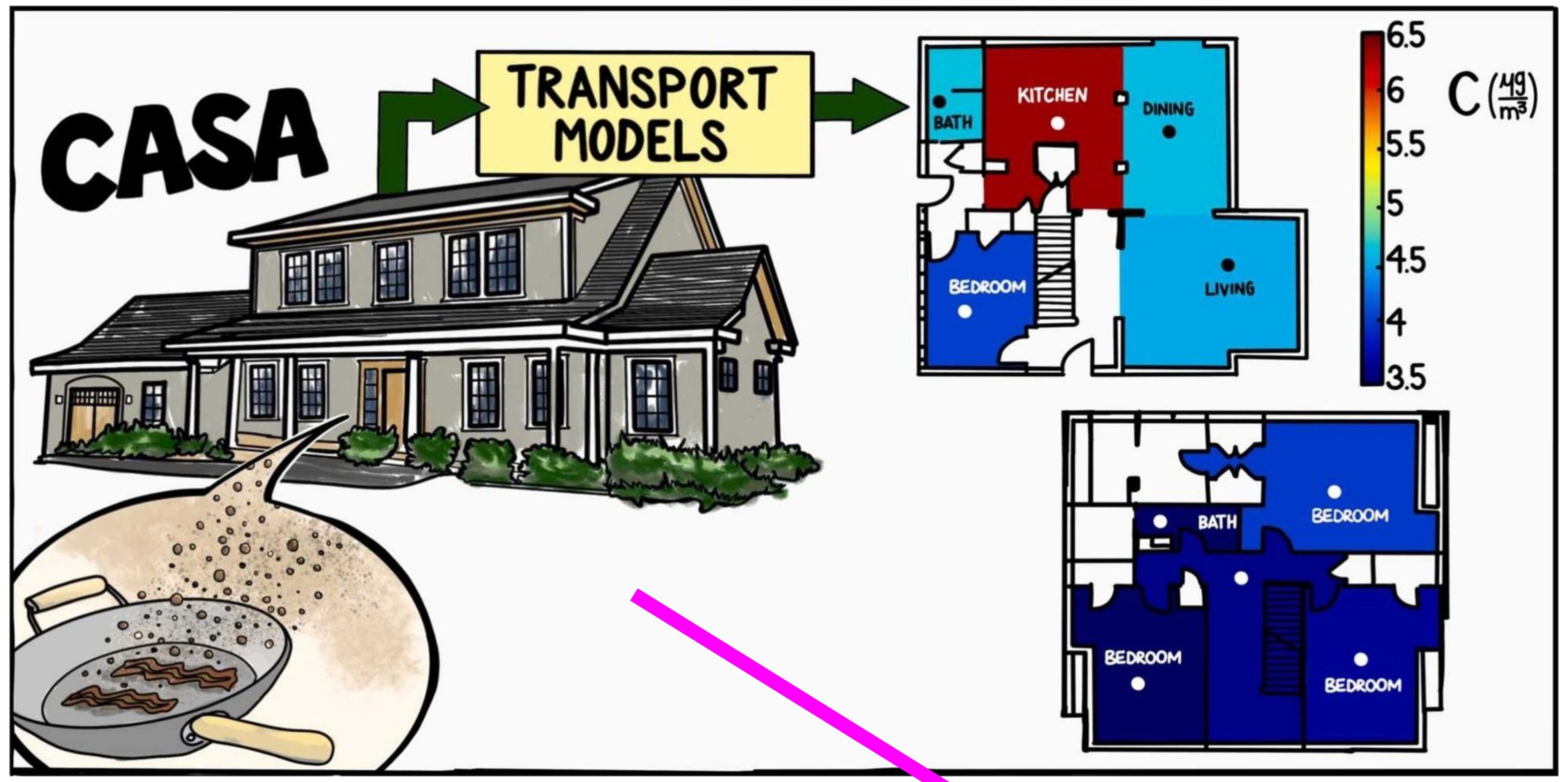
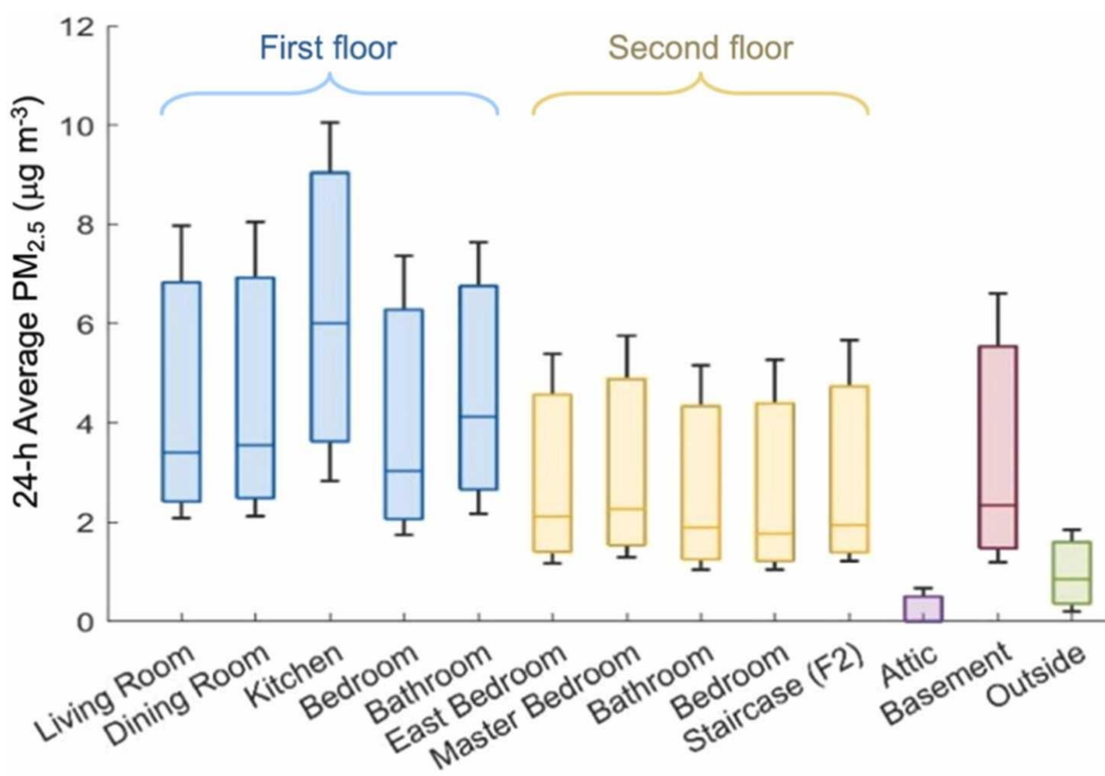
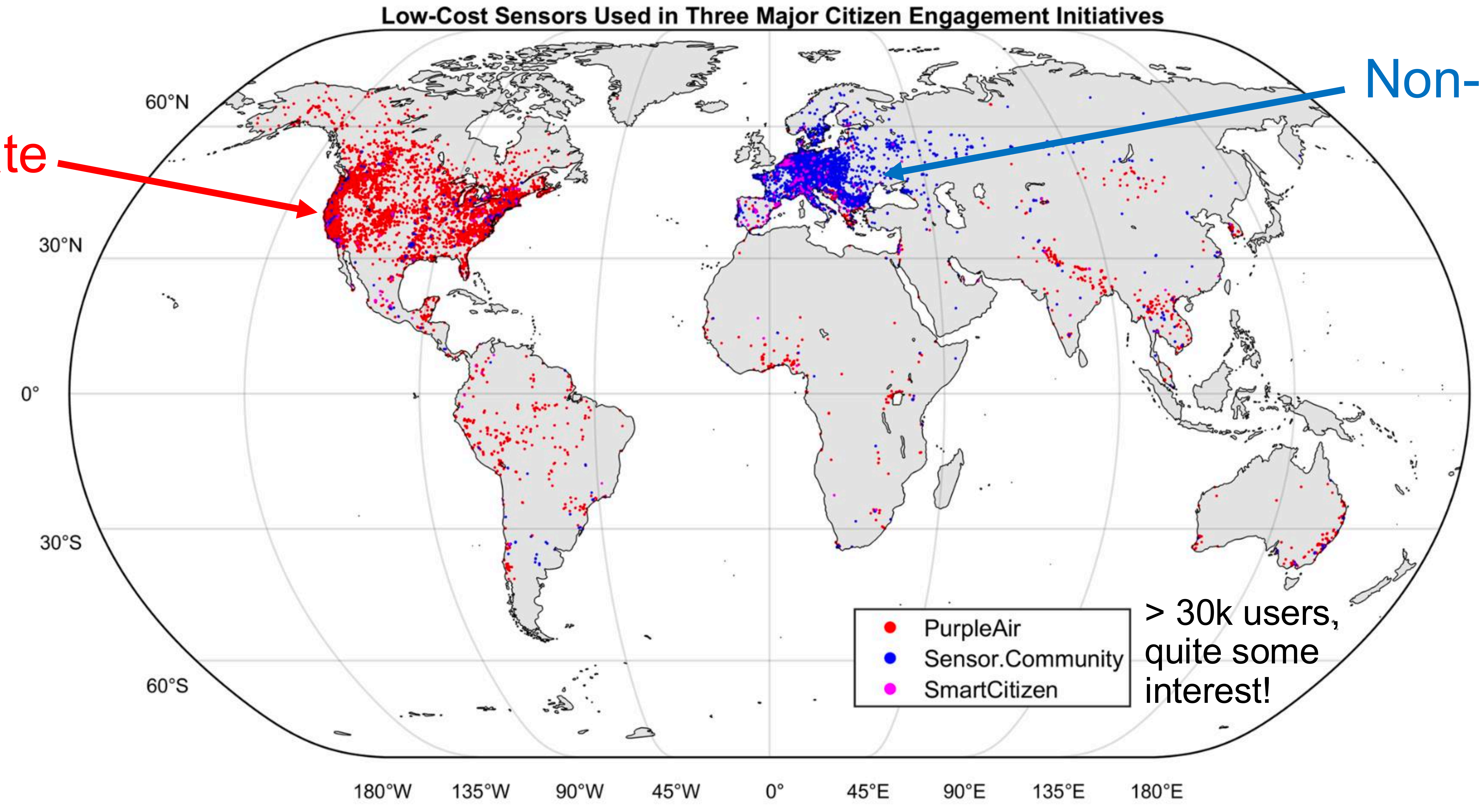




Martin, Andrew B., et al. "Investigating transport of particulate matter from cooking emissions in a multi-story house using low-cost sensor measurements and different modeling approaches." Indoor Environments (2025): 100126.

e.g., using Purple Air sensors

Existing Monitoring Tools



Private

Non-Profit

> 30k users, quite some interest!

Source: npj Climate and Atmospheric Science | (2025)8:335

We Made Our Own Tool

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Device/s

Indoor/Outdoor

Indoor

MVP
- Device Firmware ✓

Next Steps
- Commercial Device Integration trough HomeAssistant (~6 months)
- Integration with Industry Partner Building Automation System (~1 yr)

data

User Interface/s

Device GUI – WiFi AP

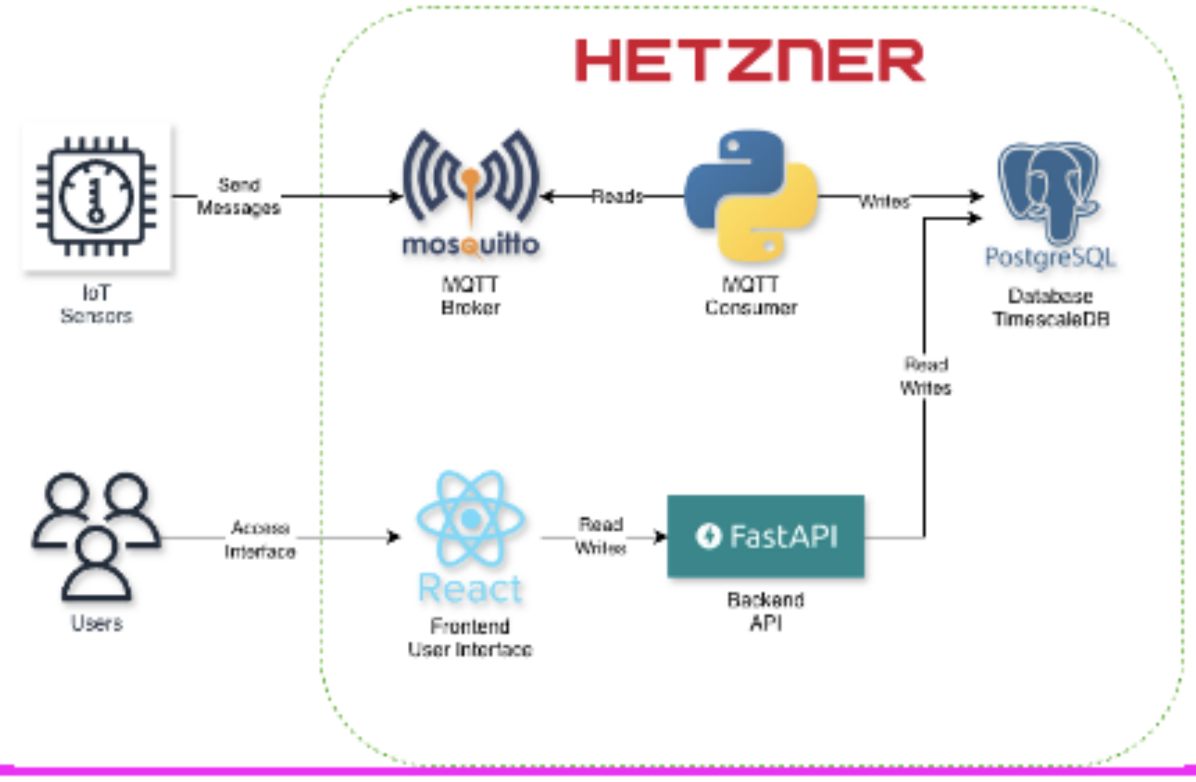
"One-time" GUI ✓
- Device Configuration
- Local Raw-Data Visualization (debug)

Web App

User Application ✓
- User Registration
- Sensor Networks Configuration
- Data Visualization / Download

Next Steps
- Improve Design

Data Management

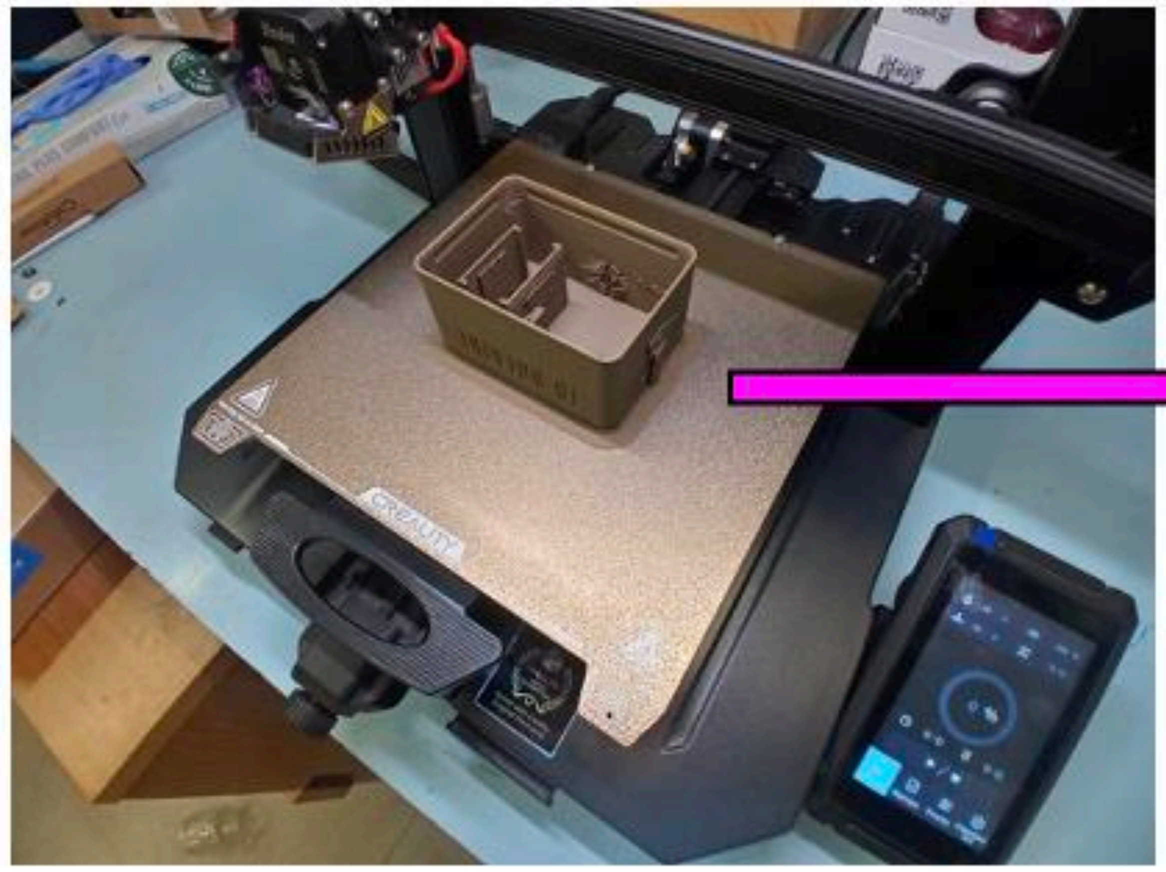


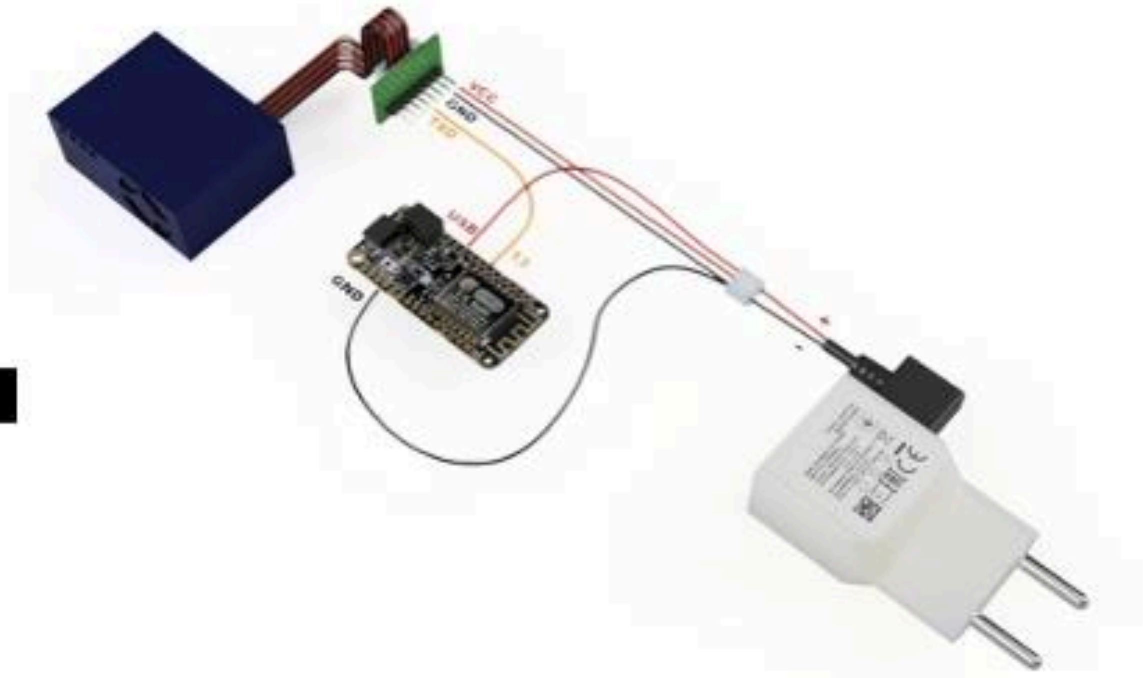
Core Application ✓
- Device Registration
- Data Handling
- Backend

Next Steps
- Mass Balance Model Integration

Contributors: Lorenzo Tenti and Alessandro Palo

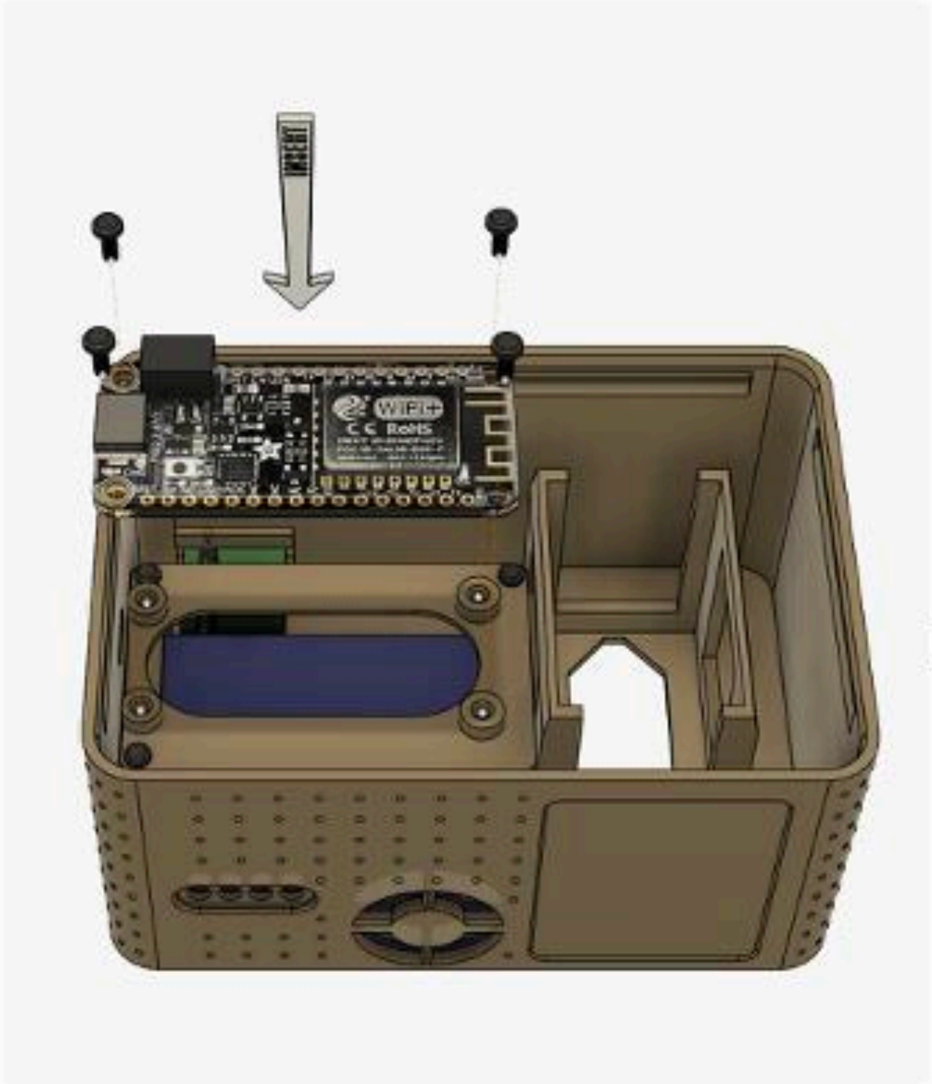
DIY Device - Hardware

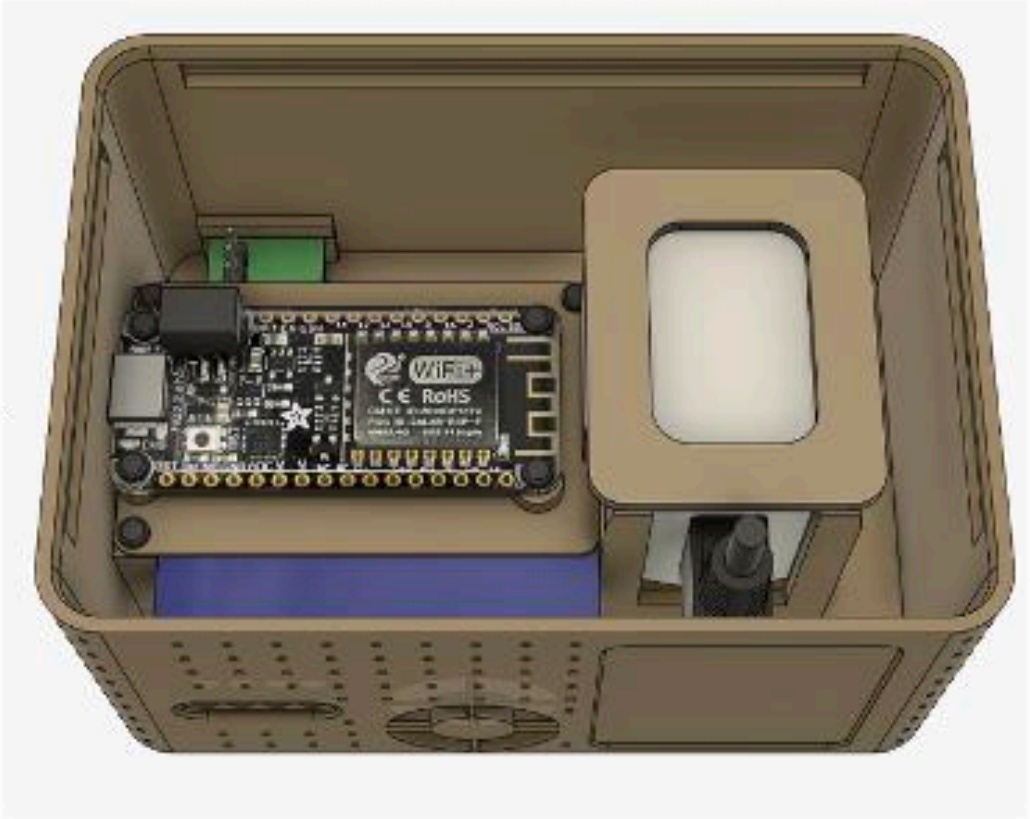




Building Materials:
- 3D printed enclosure ←
- ESP microcontroller
- Plantower PMS5003
- bme280 T-RH-P
- micro-USB power supply

→ soon: Senseair CO_2 sensor





STL file

Download STL File for 3D printer: https://makerworld.com/en/models/1912756-particulair-in-1

DIY Device - Firmware

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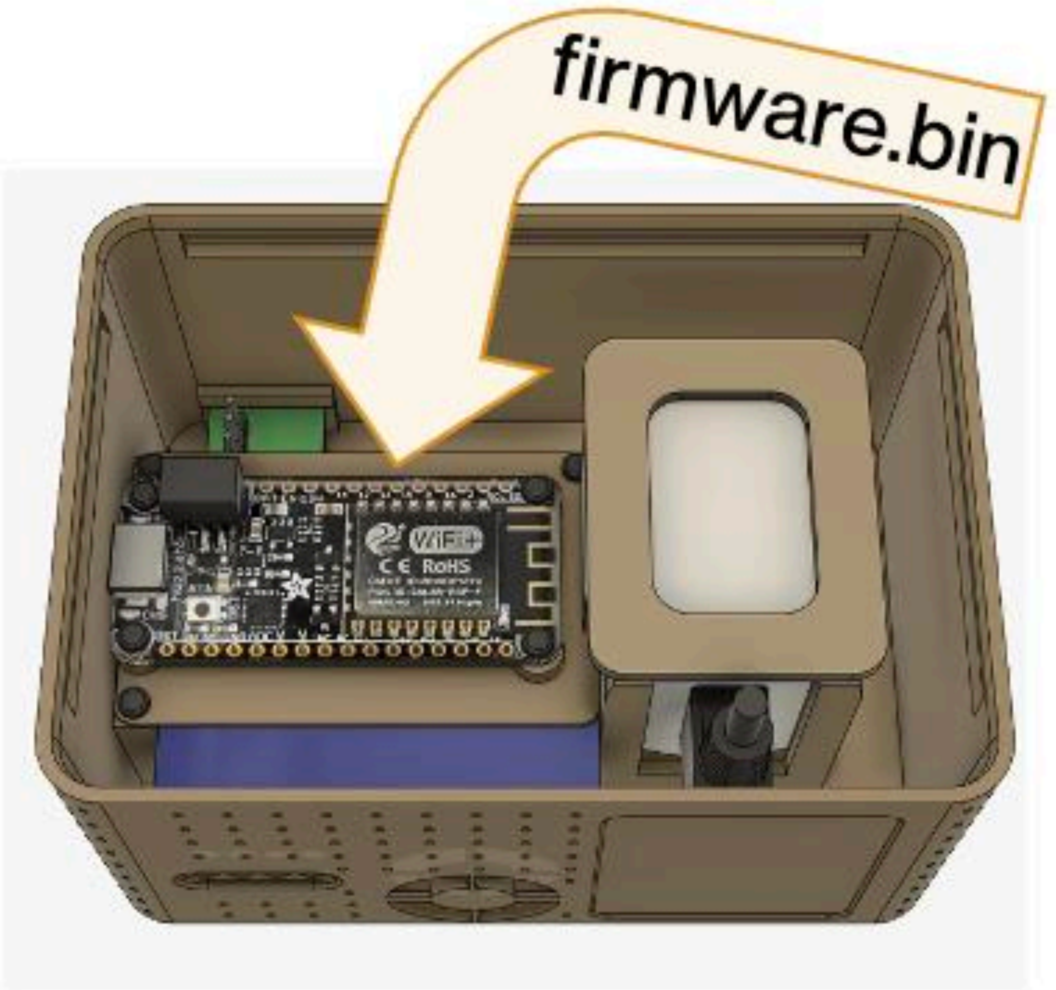
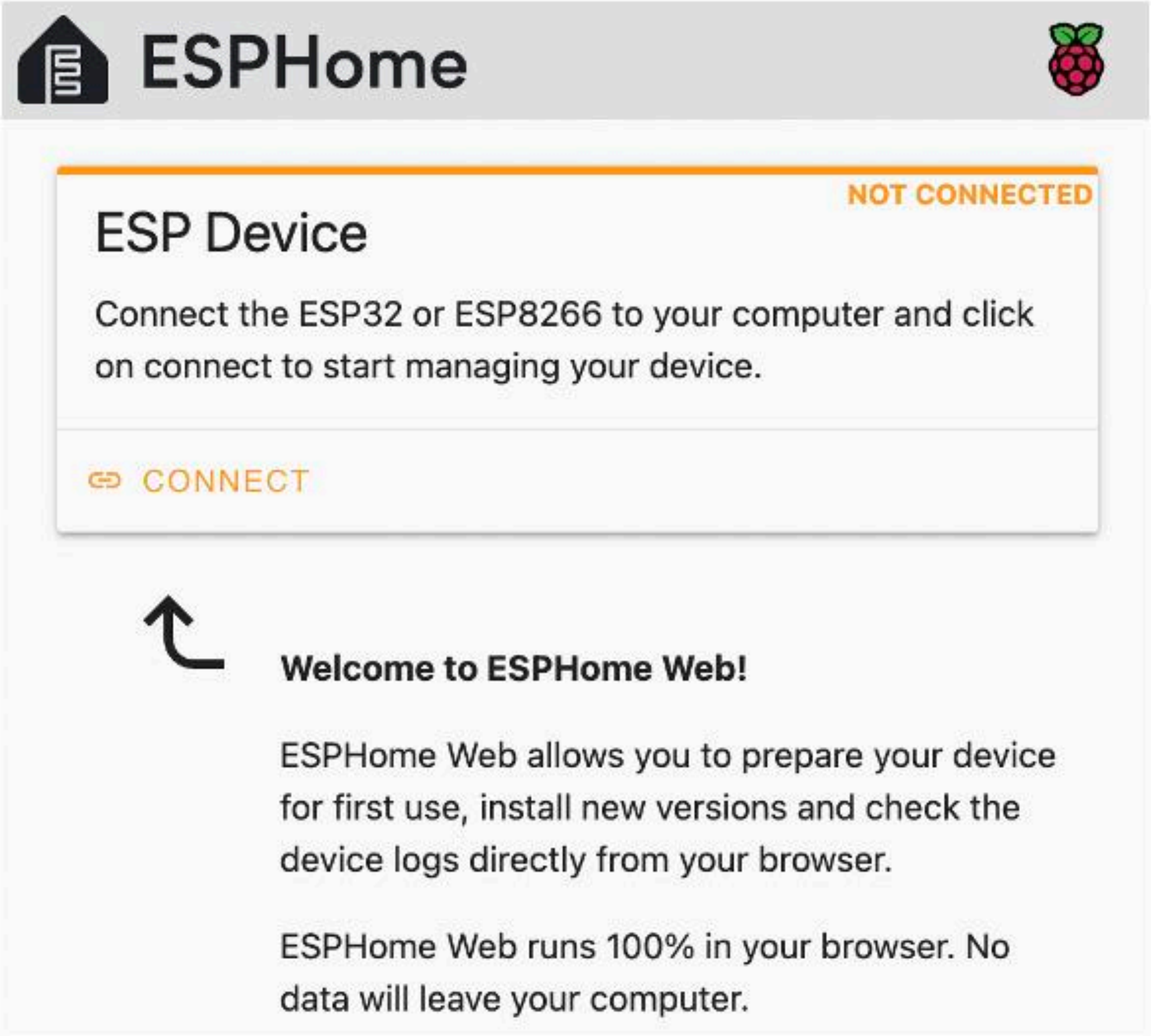
1) GitHub →



*the firmware is a file that contains the instructions for the microcontroller

2) GoTo: https://web.esphome.io/



Download "firmware.bin" for ESP8266: https://github.com/.../src/build/esp8266/firmware.bin

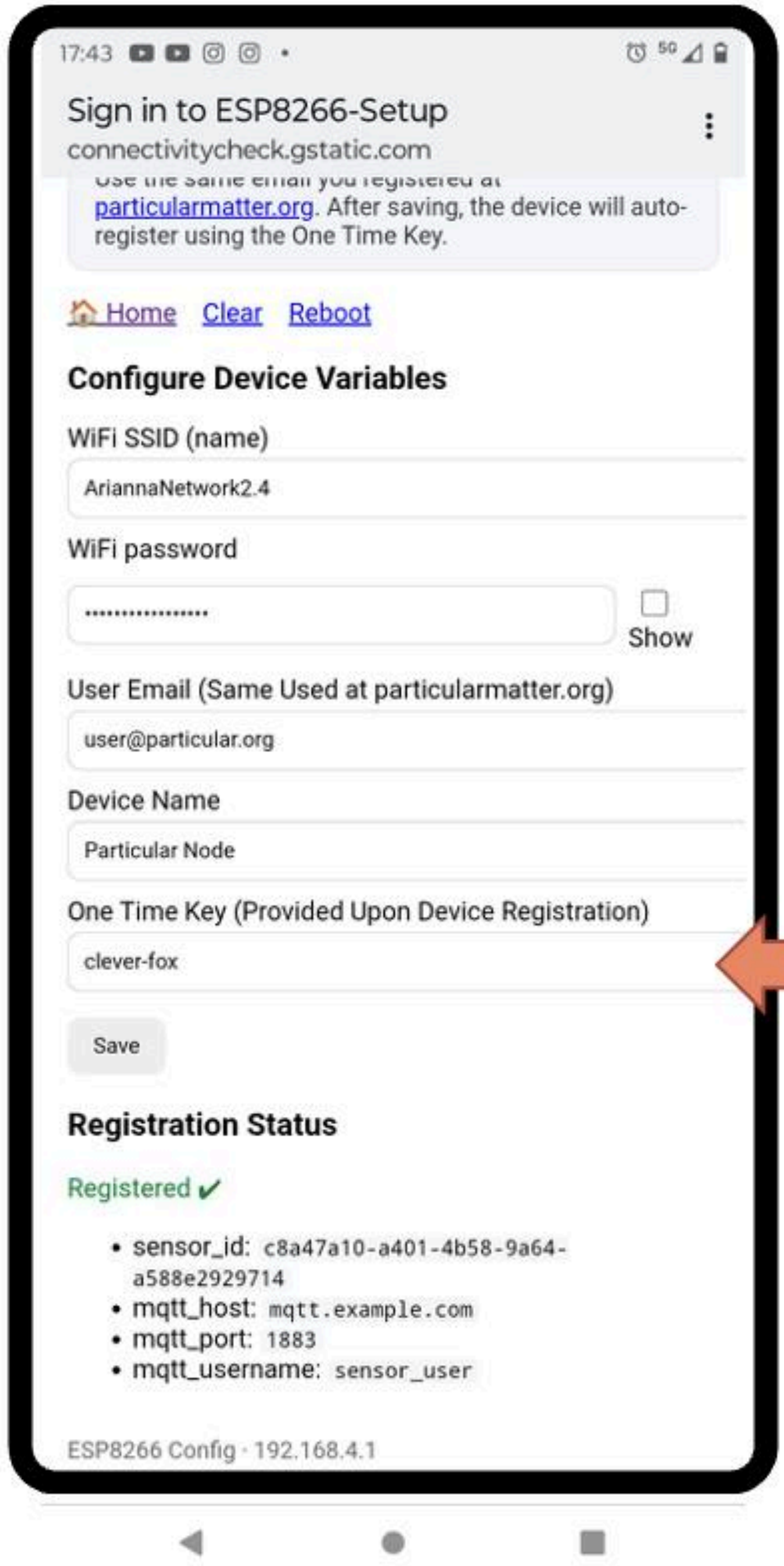
*No need to Install any Software

DIY Device - Registration

1) Plug the device to connect to the WiFi Access Point:

SSID: ParticularMatterDevice-Setup
PWD: particularmatter

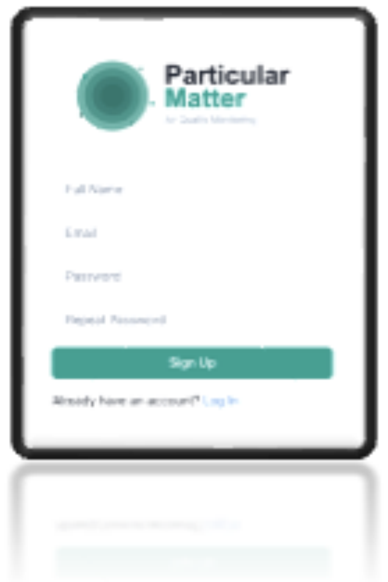
Automatic Redirect (e.g., like airports' captive portals)

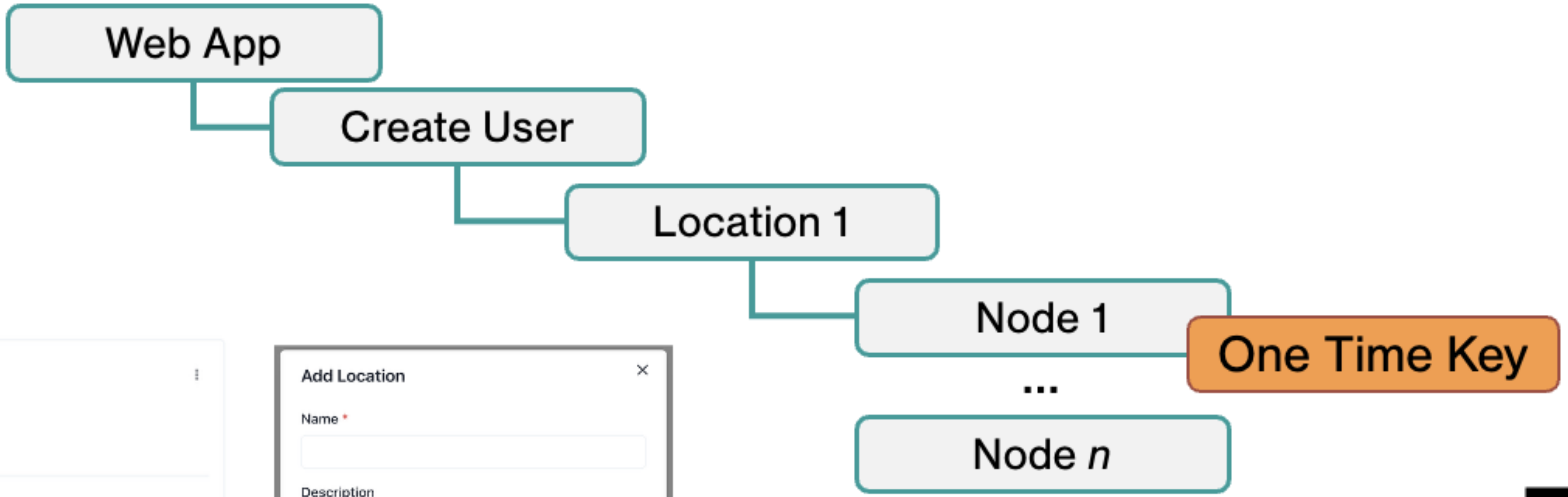


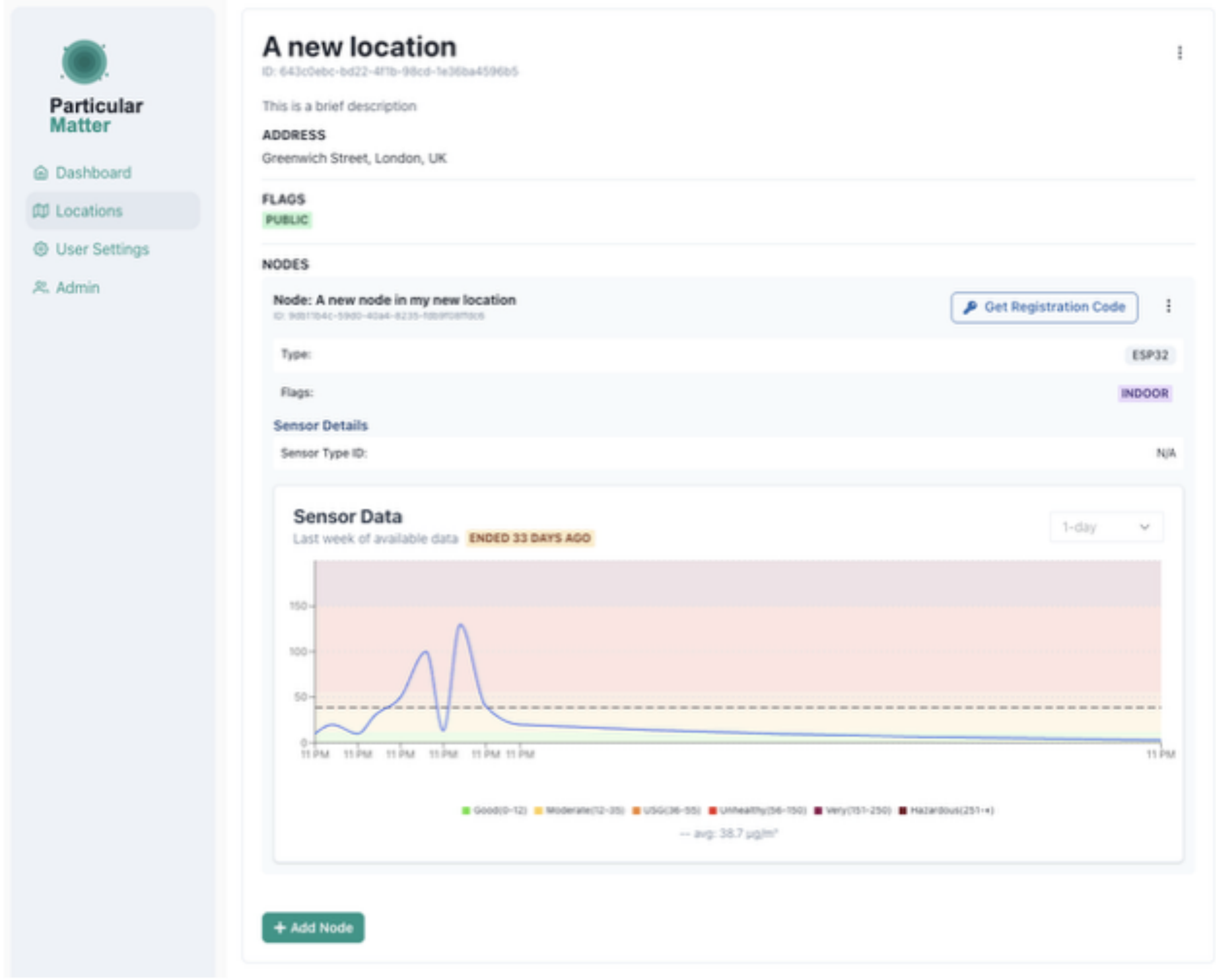
The One Time Key is generated in the Web App by the User

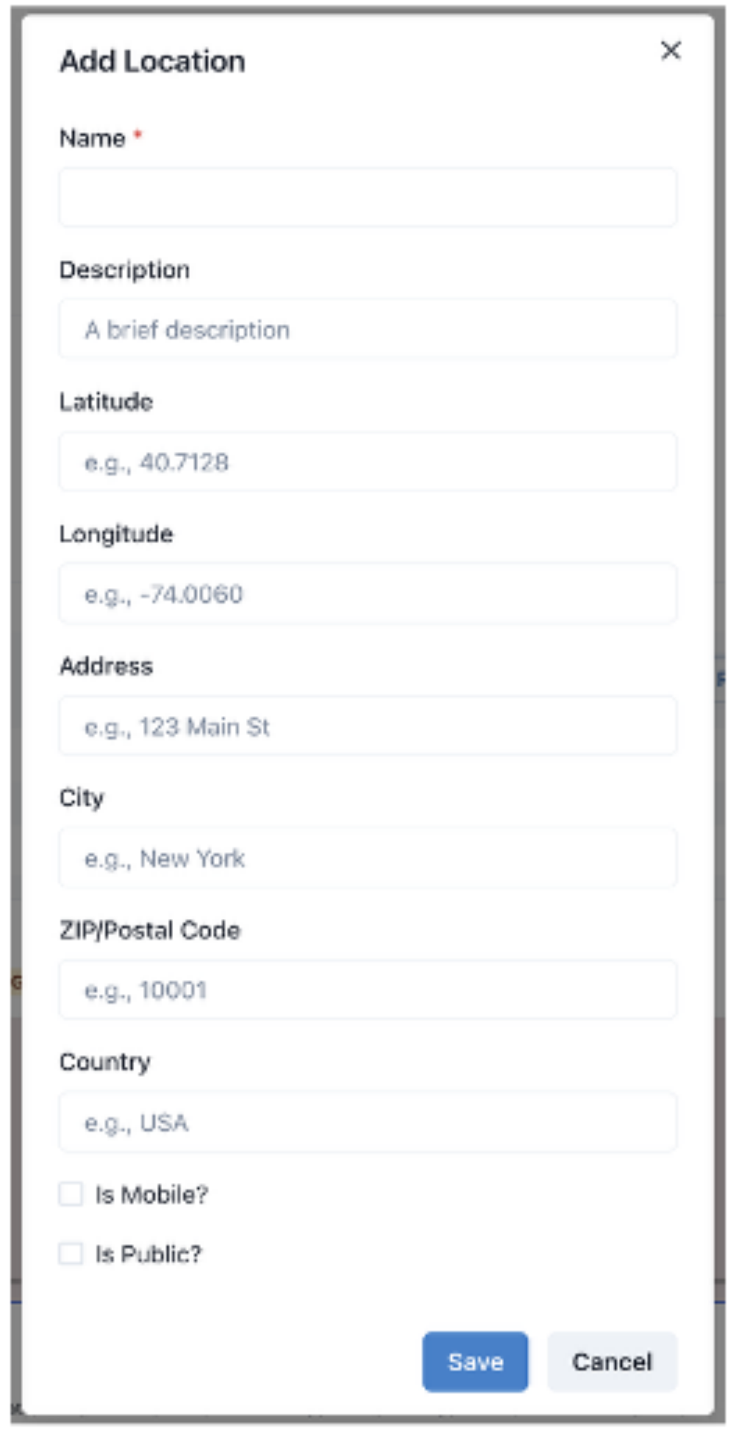
(next slide)

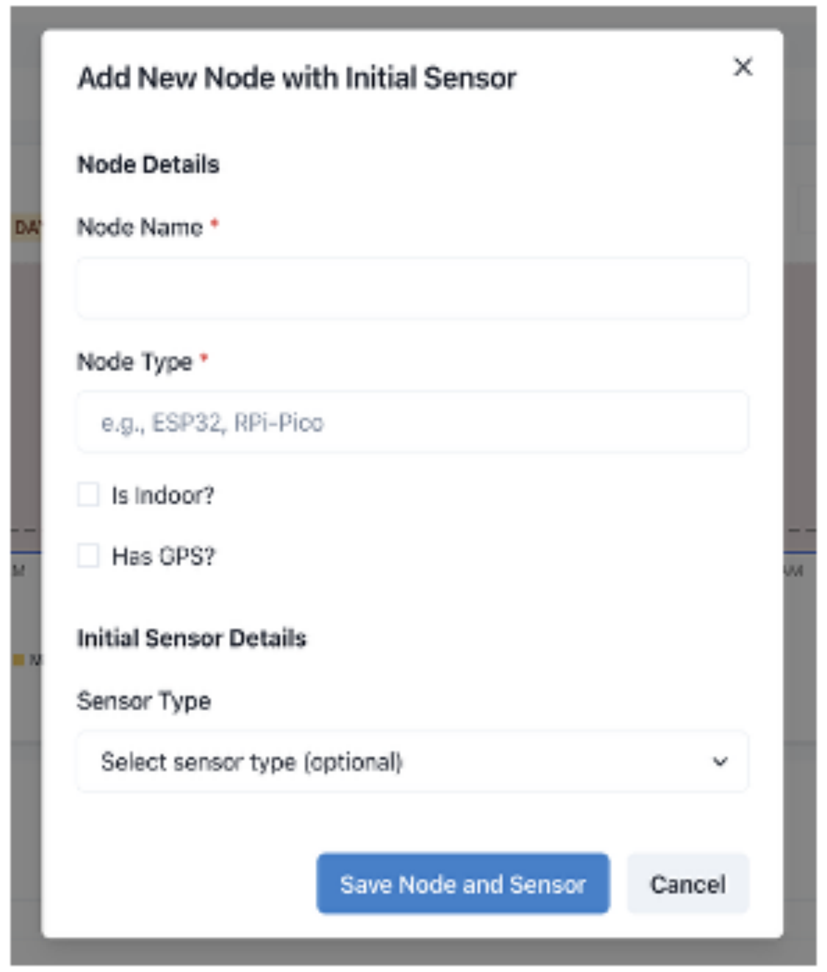
Web App – Sensor Networks Configuration

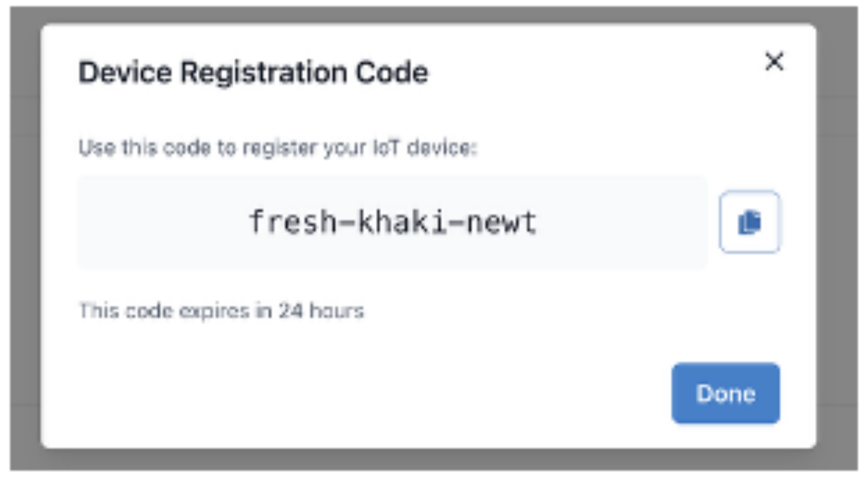












The Key Feature

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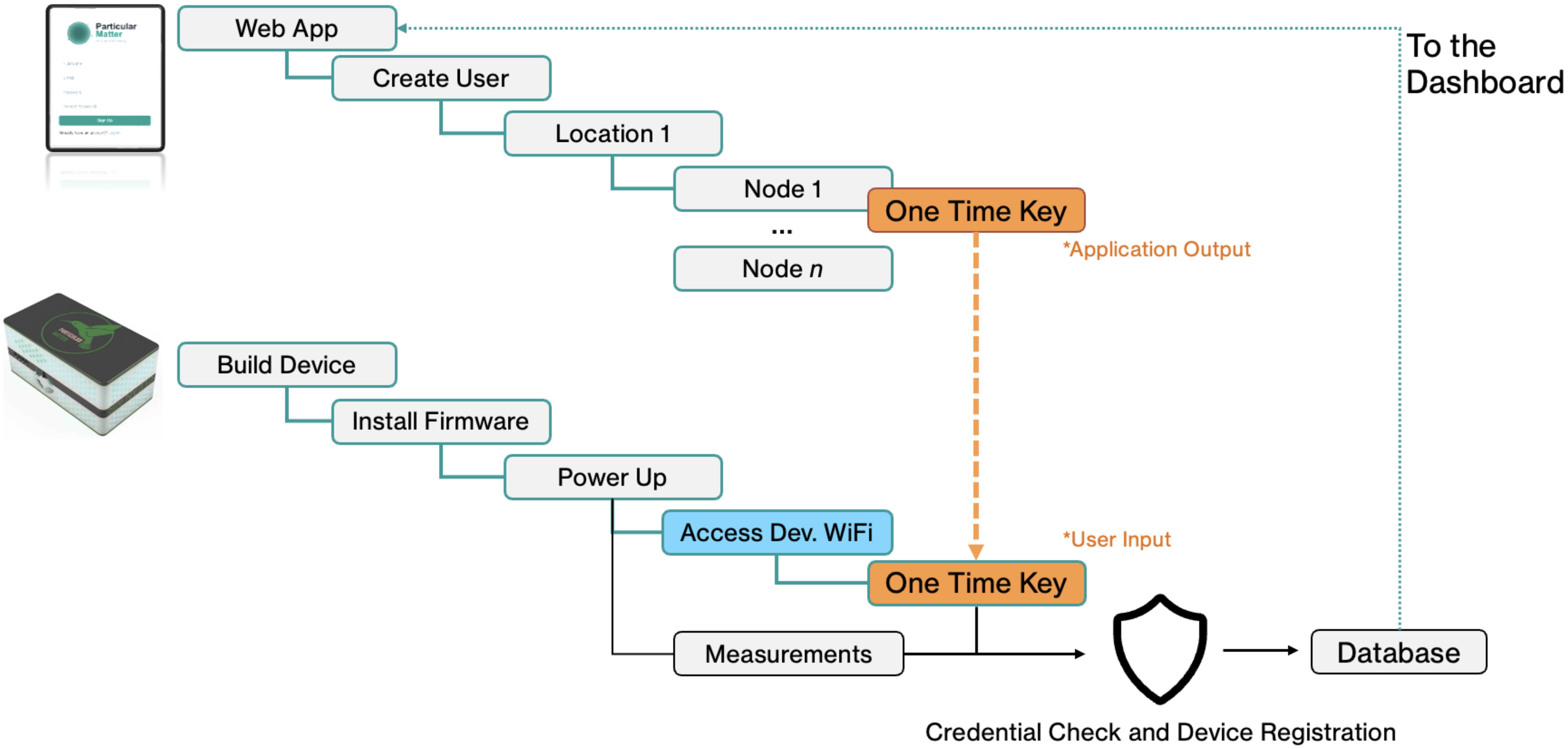
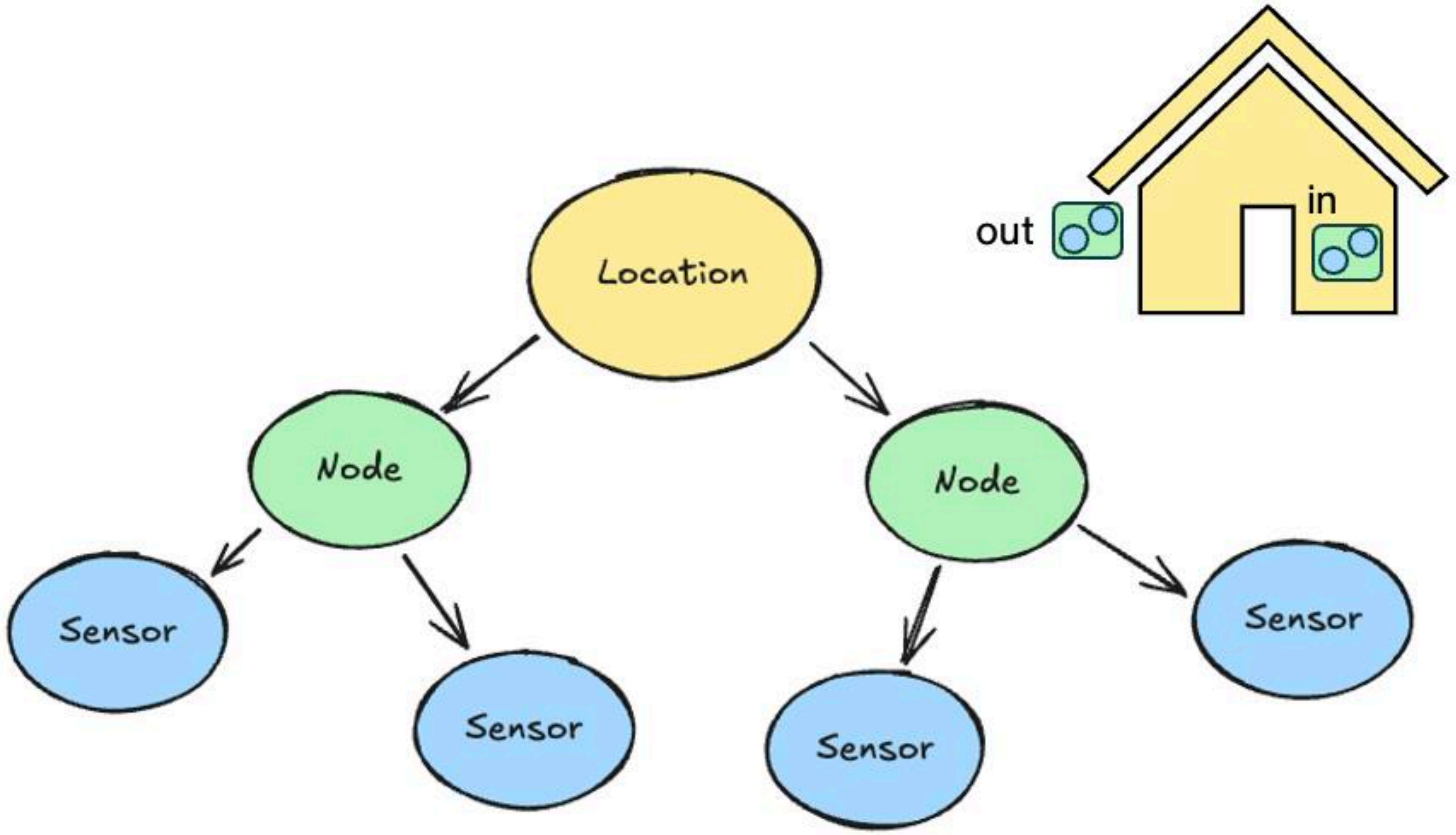
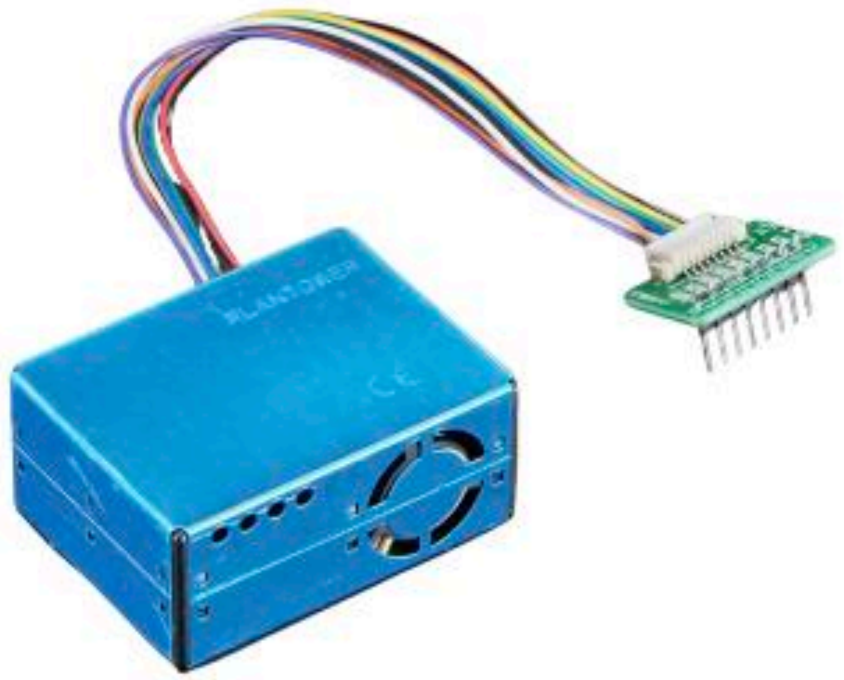
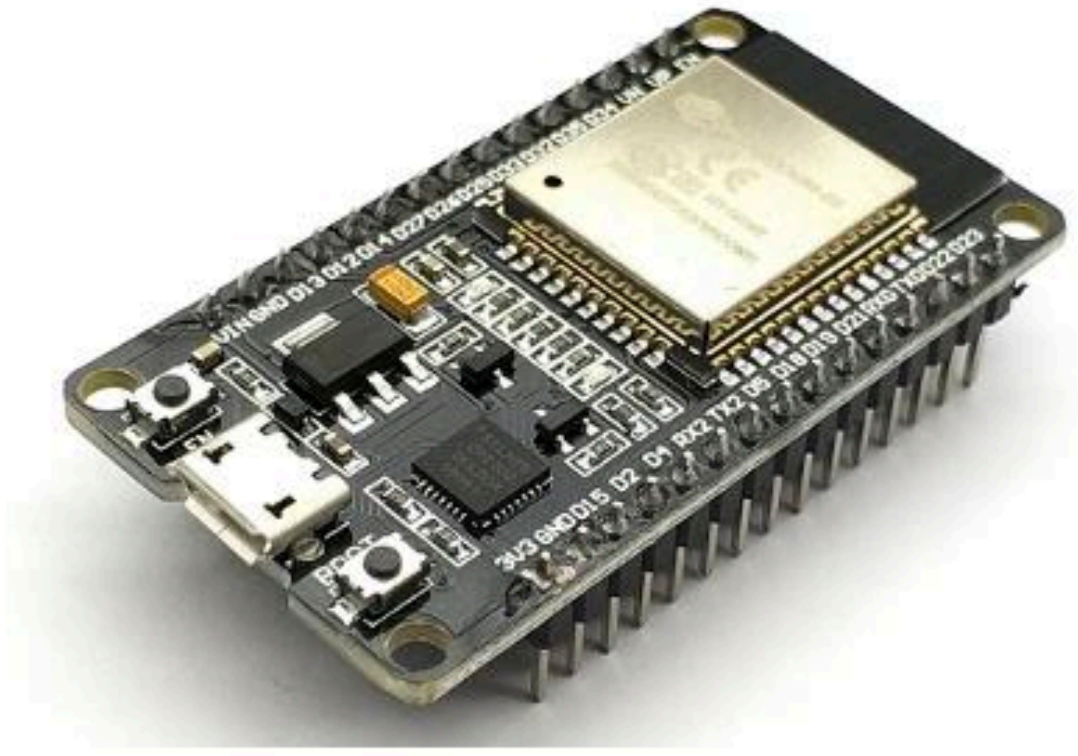


Locations & Sensors Management

- **Locations**
  - Represent physical places (e.g. home, school, park)
  - Group multiple devices under one area
  - Store GPS coordinates, address, and metadata
- **Nodes (Devices)**
  - Each *Node* = one physical device (ESP8266 or similar)
  - Connects to Wi-Fi and sends data to the platform
  - Identified by a unique registration code
- **Sensors**
  - The actual measurement components attached to a node
  - Examples: PM2.5, PM10, temperature, humidity, pressure
  - Each sensor produces time-series data
  - Easily configurable and replaceable







Web App - Main Dashboard

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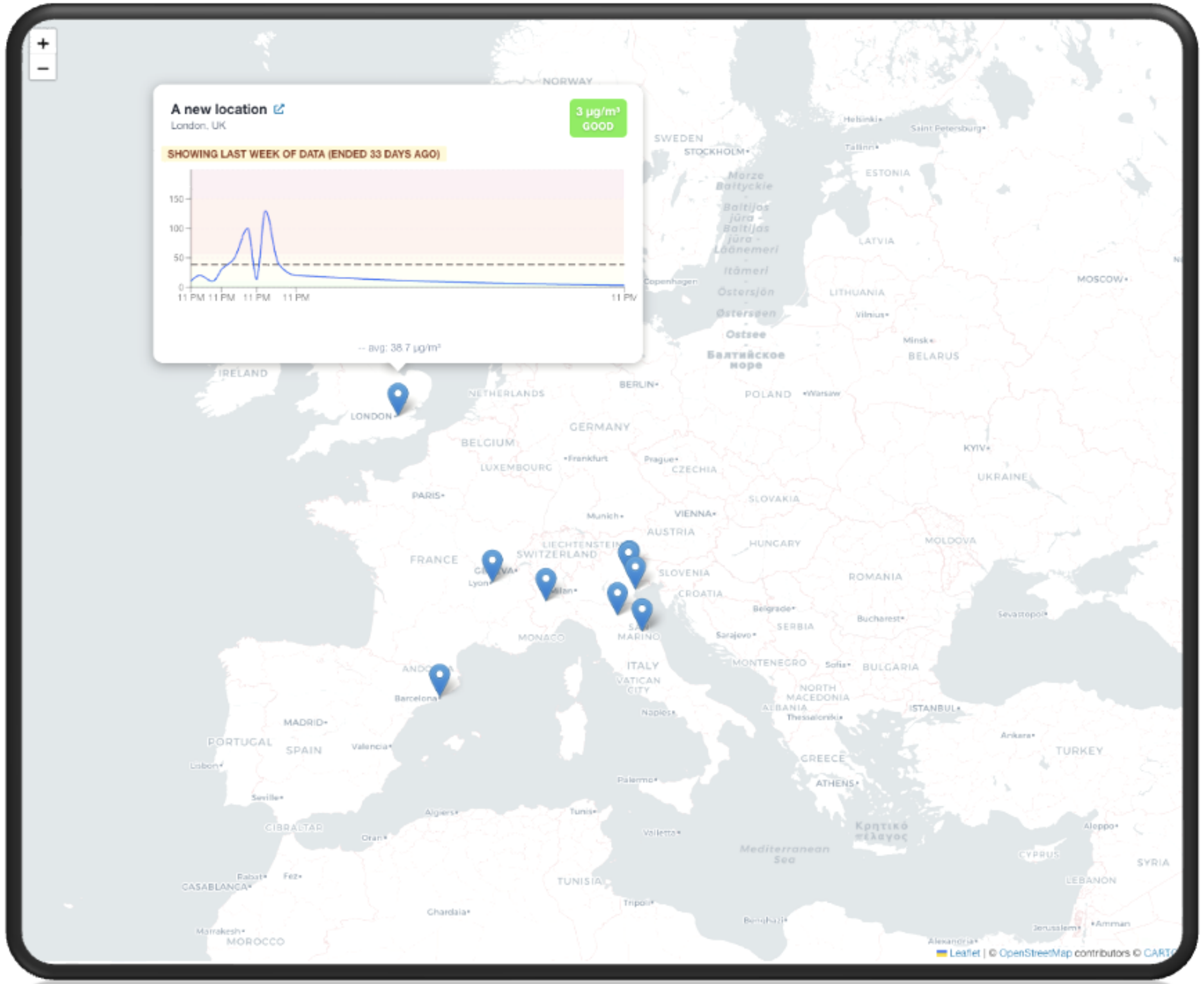
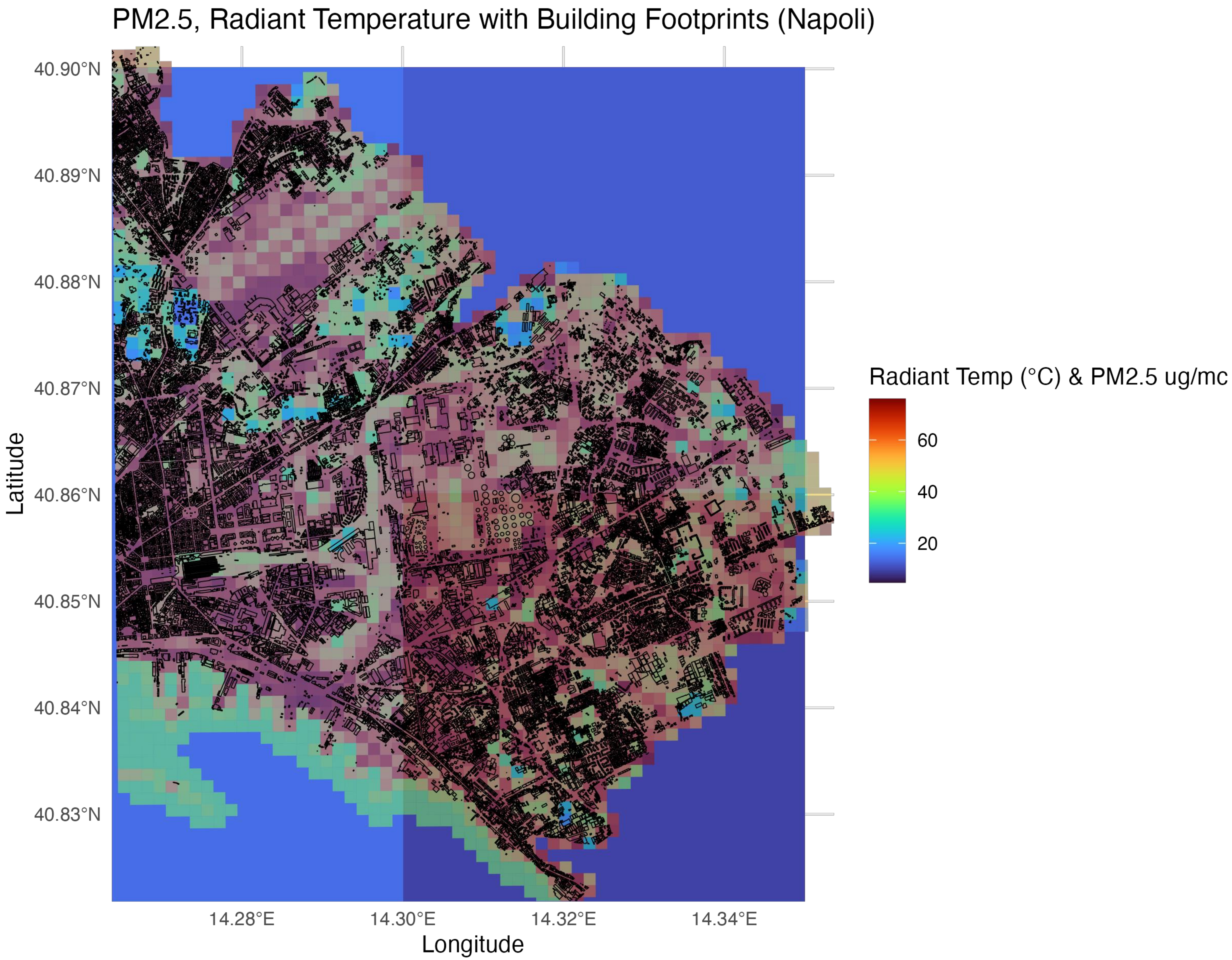
- **Explore and Visualise Data**
  - View all sensors at a glance on an interactive map (clustered or individual markers).
  - See both your private locations' data and all publicly available data.
  - Click on a marker to open a popup with live sensor readings (e.g. PM2.5, temperature, humidity).
- **Access Analytics Quickly**
  - From the map, navigate directly to a location or sensor's detail page with historical charts.
  - View aggregated air quality indicators (averages, AQI, etc.) for visible areas or selected devices.



healthRiskADAPT – Napoli as Case Study

AQ and Building Data

- estimate infiltration
- estimate behaviour
- estimate I/O ratio of PM2.5
- health impact
- …
- Simulation and Interventions
  - Urban Green
  - Technical

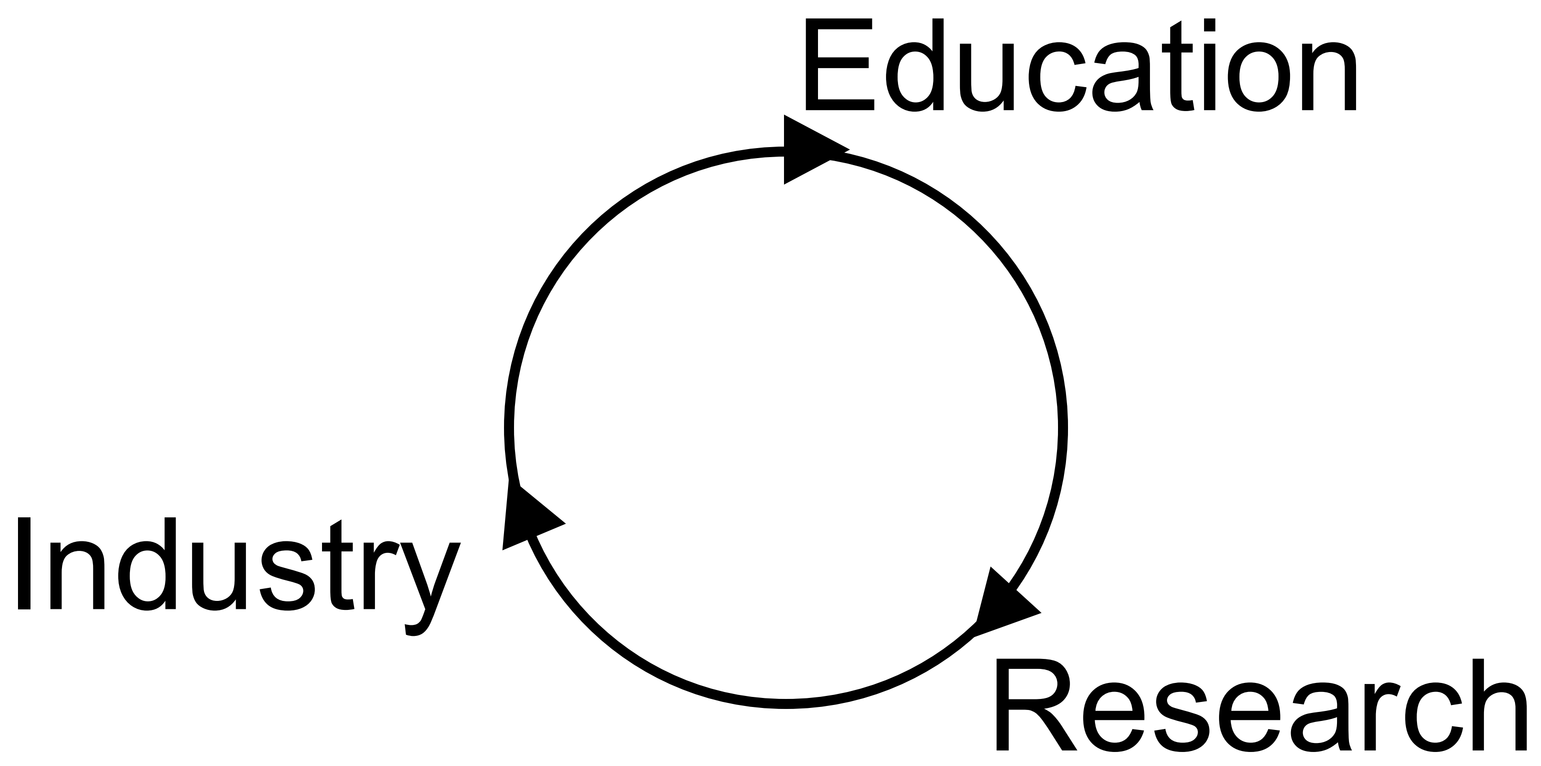


Sources:

ENEA - forAIR-it

UNINA & UCCRN - Database GIS del centro studi PLINIVS per simulazioni su isola di calore urbana e consumi energetici a Napoli Est, include edifici e copertura del suolo parametrizzati per modellazioni climatiche ed energetiche, con output su griglia 125×125 m e scenari di ondate di calore (Tmrt) e consumi 2022.

Spin-Off CNR - ParticularMatter



2020 – Technical high-school project

Final Thoughts…

Why this is relevant also for remote areas?

Cities: ~4% of EU landcover[1]

- 75% population → 85% by 2050[2]
- Consumption > 65% of global energy[3]
- 70% CO_2 emissions globally[3]
- → Environmental co-benefits are huge

EU Missions:

- Climate Resilience
- Climate-Neutral Cities

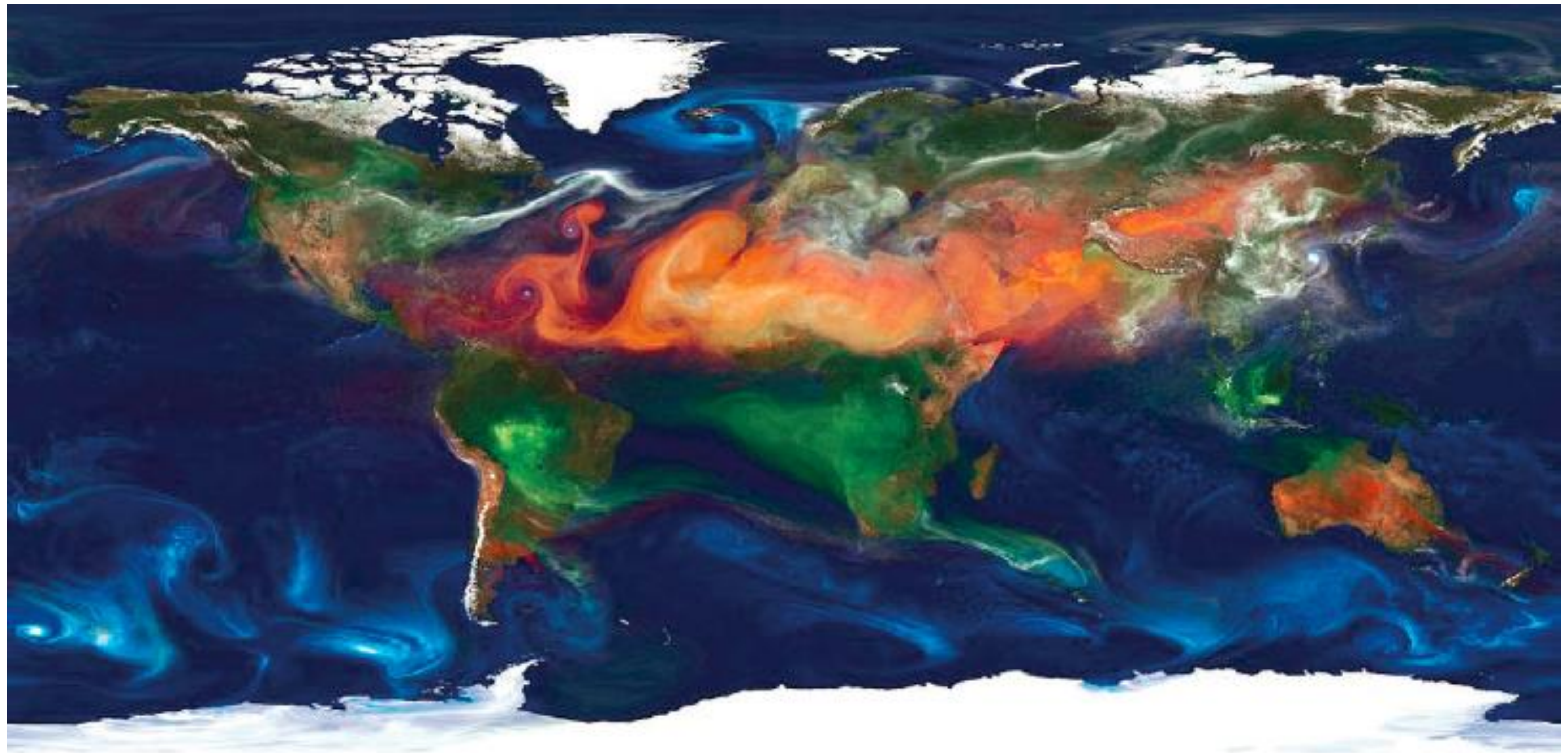


FIGURE 2-6 Output from a NASA model showing the different sources of particulate matter. Red = dust, blue = sea salt, green = smoke, and white = sulfate. SOURCE: Farmer slide 3 (NASA Center for Climate Simulation at Goddard Space Flight Center).

National Academies of Sciences, Engineering, and Medicine. 2022. Indoor Exposure to Fine Particulate Matter and Practical Mitigation Approaches: Proceedings of a Workshop. Washington, DC: The National Academies Press. https://doi.org/10.17226/26331.

Spataro, Francesca, et al. "Tracing emerging contaminants in the Arctic cryosphere: Insights from Spitsbergen (Svalbard Archipelago)." Journal of Hazardous Materials (2025): 140051.

[1]Eurostat, [2]EU Innovating cities, [3] Climate-Neutral and Smart Cities

Recap

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Environmental Monitoring

- Low-Cost Sensors
- LCS Data Quality

Built Environment and Air Pollution

- Modelling Infiltration of Outdoor Air Pollutants Indoors
- Reducing Indoor Exposure

healthRiskADAPT project

- Tools Development for Increasing Resilience to Climate Stressors Including Technological Interventions at Building Level
- → Education (Gamification) → Identify the right approach to influence **behavior** toward building more **resilient** and **sustainable societies** in the context of climate change

Q&A

Consiglio Nazionale delle Ricerche

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& Yocelyn Guerrero

Thanks!